Supplementary materials for *Estimation and inference for step-function selection models in meta-analysis with dependent effects*

Contents

A Score and Hessian of the step-function marginal log likelihood

A.1 Score functions

From Equation (7), the log of the marginal likelihood contribution for effect size estimate i from study j is given by

$$l_{ij}^{M}\left(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}\right) = \log w\left(y_{ij}, \sigma_{ij}; \boldsymbol{\zeta}\right) - \log A\left(\mathbf{x}_{ij}, \sigma_{ij}; \boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}\right) \\ - \frac{1}{2}\frac{\left(y_{ij} - \mathbf{x}_{ij}\boldsymbol{\beta}\right)^2}{\exp(\gamma) + \sigma_{ij}^2} - \frac{1}{2}\log\left(\exp(\gamma) + \sigma_{ij}^2\right) - \frac{1}{2}\log(2\pi), \tag{A1}$$

and the contribution of the estimates from study j is

$$l_{j}^{M}\left(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}\right) = \sum_{i=1}^{k_j} a_{ij} l_{ij}^{M}\left(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}\right). \tag{A2}$$

The components of the score contribution of study j involve derivatives of $l_j^M\left(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}\right)$ with respect to all model parameters, as given by

$$\mathbf{S}_j\left(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}\right) = \left(\mathbf{S}_{\beta j}', S_{\gamma j}, \mathbf{S}_{\zeta j}'\right)'. \tag{A3}$$

Let $w_{ij} = w\left(y_{ij}, \sigma_{ij}; \boldsymbol{\lambda}\right)$, $\mu_{ij} = \mathbf{x}_{ij}\boldsymbol{\beta}$, $\eta_{ij} = \exp(\gamma) + \sigma_{ij}^2$, and $A_{ij} = A\left(\mathbf{x}_{ij}, \sigma_{ij}; \boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}\right)$ as defined in Equation (6). The score contribution of study j for the meta-regression coefficients has the general form

$$\mathbf{S}_{\beta j} = \sum_{i=1}^{k_j} a_{ij}\mathbf{x}_{ij}'\left(\frac{\left(y_{ij} - \mu_{ij}\right)}{\eta_{ij}} - \frac{1}{A_{ij}}\frac{\partial A_{ij}}{\partial \mu_{ij}}\right). \tag{A4}$$

The score for the variance parameter has the general form

$$S_{\gamma j} = \sum_{i=1}^{k_j} a_{ij}\tau^2\left(\frac{\left(y_{ij} - \mu_{ij}\right)^2 - \eta_{ij}}{2\eta_{ij}^2} - \frac{1}{A_{ij}}\frac{\partial A_{ij}}{\partial \eta_{ij}}\right). \tag{A5}$$

The score for the selection parameters has the general form $\mathbf{S}_{\boldsymbol{\zeta} j} = (S_{\zeta_1 j}, ..., S_{\zeta_H j})'$, where

$$S_{\zeta_h j} = \sum_{i=1}^{k_j} a_{ij} \lambda_h \left(\frac{1}{w_{ij}} \frac{\partial w_{ij}}{\partial \lambda_h} - \frac{1}{A_{ij}} \frac{\partial A_{ij}}{\partial \lambda_h} \right), \tag{A6}$$

for $h = 1, ..., H$.

For the step-function selection model, the derivatives of w_{ij} and A_{ij} are given by

$$\frac{\partial w_{ij}}{\partial \lambda_h} = I\left(\alpha_h < p_{ij} \leq \alpha_{h+1}\right) \tag{A7}$$

$$\frac{\partial A_{ij}}{\partial \mu_{ij}} = \frac{1}{\eta_{ij}^{1/2}} \sum_{h=0}^{H} \lambda_h \left[\phi\left(c_{h+1,ij}\right) - \phi\left(c_{hij}\right)\right] \tag{A8}$$

$$\frac{\partial A_{ij}}{\partial \eta_{ij}} = \frac{1}{2\eta_{ij}} \sum_{h=0}^{H} \lambda_h \left[c_{h+1,ij} \phi\left(c_{h+1,ij}\right) - c_{hij} \phi\left(c_{hij}\right)\right] \tag{A9}$$

$$\frac{\partial A_{ij}}{\partial \lambda_h} = B_{hij}, \tag{A10}$$

with $c_{hij} = \left[\sigma_{ij} \Phi^{-1}\left(1 - \alpha_h\right) - \mathbf{x}_{ij} \boldsymbol{\beta}\right] / \sqrt{\tau^2 + \sigma_{ij}^2}$ and $B_{hij} = \Phi\left(c_{hij}\right) - \Phi\left(c_{h+1,ij}\right)$ for $h = 1, ..., H$.

A.2 Hessian

The sub-components of the Hessian matrix from study j have the following forms:

$$\mathbf{H}_j^{\beta\beta'} = \frac{\partial}{\partial \boldsymbol{\beta}'}\mathbf{S}_{\beta j} = \sum_{i=1}^{k_j} a_{ij}\mathbf{x}_{ij}'\mathbf{x}_{ij}\left(\frac{1}{A_{ij}^2}\left(\frac{\partial A_{ij}}{\partial \mu_{ij}}\right)^2 - \frac{1}{A_{ij}}\frac{\partial^2 A_{ij}}{\partial \mu_{ij}^2} - \frac{1}{\eta_{ij}}\right) \tag{A11}$$

$$\mathbf{H}_j^{\beta\gamma} = \frac{\partial}{\partial \gamma}\mathbf{S}_{\beta j} = \sum_{i=1}^{k_j} a_{ij}\mathbf{x}_{ij}'\tau^2\left(\frac{1}{A_{ij}^2}\frac{\partial A_{ij}}{\partial \mu_{ij}}\frac{\partial A_{ij}}{\partial \eta_{ij}} - \frac{1}{A_{ij}}\frac{\partial^2 A_{ij}}{\partial \mu_{ij}\partial \eta_{ij}} - \frac{y_{ij}-\mu_{ij}}{\eta_{ij}^2}\right) \tag{A12}$$

$$\mathbf{H}_j^{\beta\zeta_h} = \frac{\partial}{\partial \zeta_h}\mathbf{S}_{\beta j} = \sum_{i=1}^{k_j} a_{ij}\mathbf{x}_{ij}'\lambda_h\left(\frac{1}{A_{ij}^2}\frac{\partial A_{ij}}{\partial \mu_{ij}}\frac{\partial A_{ij}}{\partial \lambda_h} - \frac{1}{A_{ij}}\frac{\partial^2 A_{ij}}{\partial \mu_{ij}\partial \lambda_h}\right) \tag{A13}$$

$$\begin{aligned}\mathbf{H}_j^{\gamma\gamma} = \frac{\partial}{\partial \gamma}S_{\gamma j} = \sum_{i=1}^{k_j} a_{ij}\Bigg[\tau^2\left(\frac{(y_{ij}-\mu_{ij})^2-\eta_{ij}}{2\eta_{ij}^2} - \frac{1}{A_{ij}}\frac{\partial A_{ij}}{\partial \eta_{ij}}\right)\\ + \tau^4\left(\frac{1}{A_{ij}^2}\left(\frac{\partial A_{ij}}{\partial \eta_{ij}}\right)^2 - \frac{1}{A_{ij}}\frac{\partial^2 A_{ij}}{\partial \eta_{ij}^2} - \frac{2(y_{ij}-\mu_{ij})^2-\eta_{ij}}{2\eta_{ij}^3}\right)\Bigg]\end{aligned} \tag{A14}$$

$$\mathbf{H}_j^{\gamma\zeta_h} = \frac{\partial}{\partial \zeta_h}S_{\gamma j} = \sum_{i=1}^{k_j} a_{ij}\tau^2\lambda_h\left(\frac{1}{A_{ij}^2}\frac{\partial A_{ij}}{\partial \eta_{ij}}\frac{\partial A_{ij}}{\partial \lambda_h} - \frac{1}{A_{ij}}\frac{\partial^2 A_{ij}}{\partial \eta_{ij}\partial \lambda_h}\right) \tag{A15}$$

$$\begin{aligned}\mathbf{H}_j^{\zeta_f\zeta_h} = \frac{\partial}{\partial \zeta_h}S_{\zeta_f j} = \sum_{i=1}^{k_j} a_{ij}\Bigg[\lambda_f\lambda_h\left(\frac{1}{A_{ij}^2}\frac{\partial A_{ij}}{\partial \lambda_f}\frac{\partial A_{ij}}{\partial \lambda_h} - \frac{1}{w_{ij}^2}\frac{\partial w_{ij}}{\partial \lambda_f}\frac{\partial w_{ij}}{\partial \lambda_h}\right)\\ + I(f=h)\lambda_h\left(\frac{1}{w_{ij}}\frac{\partial w_{ij}}{\partial \lambda_h} - \frac{1}{A_{ij}}\frac{\partial A_{ij}}{\partial \lambda_h}\right)\Bigg],\end{aligned} \tag{A16}$$

where the second partial derivatives of A_{ij} are given by

$$\frac{\partial^2 A_{ij}}{\partial \mu_{ij}^2} = \frac{1}{\eta_{ij}}\sum_{h=0}^{H}\lambda_h\left[c_{h+1,ij}\phi(c_{h+1,ij}) - c_{hij}\phi(c_{hij})\right] \tag{A17}$$

$$\frac{\partial^2 A_{ij}}{\partial \mu_{ij}\partial \eta_{ij}} = \frac{1}{2\eta_{ij}^{3/2}}\sum_{h=0}^{H}\lambda_h\left[(c_{h+1,ij}^2-1)\phi(c_{h+1,ij}) - (c_{hij}^2-1)\phi(c_{hij})\right] \tag{A18}$$

$$\frac{\partial^2 A_{ij}}{\partial \mu_{ij}\partial \lambda_h} = \frac{\phi\left(c_{h+1,ij}\right) - \phi\left(c_{hij}\right)}{\eta_{ij}^{1/2}} \tag{A19}$$

$$\frac{\partial^2 A_{ij}}{\partial \eta_{ij}^2} = \frac{1}{4\eta_{ij}^2}\sum_{h=0}^{H}\lambda_h\left[(c_{h+1,ij}^3-3c_{h+1,ij})\phi(c_{h+1,ij}) - (c_{hij}^3-3c_{hij})\phi(c_{hij})\right] \tag{A20}$$

$$\frac{\partial^2 A_{ij}}{\partial \eta_{ij}\partial \lambda_h} = \frac{c_{h+1,ij}\phi\left(c_{h+1,ij}\right) - c_{hij}\phi\left(c_{hij}\right)}{2\eta_{hij}}. \tag{A21}$$

A.3 Derivatives of the prior penalty functions

Equation (9) references the derivatives of the prior penalty terms specified in Equations (15), (16), and (17). Denote the full derivative vector as

$$p'\left(\hat{\boldsymbol{\beta}}, \hat{\gamma}, \hat{\boldsymbol{\zeta}}\right) = \begin{pmatrix} p'_{\beta}(\boldsymbol{\beta}) \\ p'_{\gamma}(\gamma) \\ p'_{\zeta}(\boldsymbol{\zeta}) \end{pmatrix}. \tag{A22}$$

The components of the derivative vector are

$$p'_{\beta}(\boldsymbol{\beta}) = -\frac{1}{4}\boldsymbol{\beta}^3 \tag{A23}$$

$$p'_{\gamma}(\gamma) = \frac{1}{2} - \frac{5}{2}\exp\left(\frac{\gamma}{2}\right) \tag{A24}$$

$$p'_{\zeta}(\boldsymbol{\zeta}) = -\frac{2}{27}\left(\boldsymbol{\zeta} - \log\left(\frac{1}{2}\right)\mathbf{1}_H\right)^3, \tag{A25}$$

where $\mathbf{1}_H$ denotes an $H \times 1$ vector with unit entries and powers of vector terms are evaluated element-wise.

B Technical details for the augmented, re-weighted Gaussian likelihood estimator

B.1 Estimating equations for $\boldsymbol{\beta}$ and γ

We defined the ARGL estimator as the solution to a vector of estimating equations as given in Equation (14), where the contribution of study j is

$$\mathbf{M}_j(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}) = \begin{bmatrix} \mathbf{S}^G_{\beta j}(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}) \\ S^G_{\gamma j}(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}) \\ \mathbf{S}_{\zeta j}(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}) \end{bmatrix}, \tag{B1}$$

as given in Equation (13). The first two components of $\mathbf{M}_j(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta})$ are given by

$$\mathbf{S}^G_{\beta j}(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}) = \sum_{i=1}^{k_j} a_{ij} \times \mathbf{x}'_{ij} \frac{y_{ij} - \mathbf{x}_{ij}\boldsymbol{\beta}}{w_{ij}\left(\exp(\gamma) + \sigma^2_{ij}\right)} \tag{B2}$$

$$\mathbf{S}^G_{\gamma j}(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}) = \sum_{i=1}^{k_j} a_{ij} \times \frac{\exp(\gamma)}{2w_{ij}} \left(\frac{(y_{ij} - \mathbf{x}_{ij}\boldsymbol{\beta})^2}{\left(\exp(\gamma) + \sigma^2_{ij}\right)^2} - \frac{1}{\exp(\gamma) + \sigma^2_{ij}} \right). \tag{B3}$$

The third component is the score equation for $\boldsymbol{\zeta}$ based on the marginal likelihood, as given in Equation (A6).

B.2 Sandwich estimator for the augmented, re-weighted Gaussian likelihood estimator

The augmented, re-weighted Gaussian likelihood (ARGL) estimators are defined as the solution to the estimating equations given in Equation (13) and (14). Sandwich variance approximations for the ARGL estimators involve the estimating equations and their Jacobian.

Let $\mathbf{J}$ denote the Jacobian matrix of the ARGL estimating equations, given by

$$\mathbf{J}(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta}) = \sum_{j=1}^{J} \frac{\partial \mathbf{M}_j(\boldsymbol{\beta}, \gamma, \boldsymbol{\zeta})}{\partial(\boldsymbol{\beta}' \gamma \boldsymbol{\zeta}')}. \tag{B4}$$

The exact form of the Jacobian is described below. Let $\tilde{\mathbf{M}}_j = \mathbf{M}_j(\tilde{\boldsymbol{\beta}}, \tilde{\gamma}, \tilde{\boldsymbol{\zeta}})$ and $\tilde{\mathbf{J}} = \mathbf{J}(\tilde{\boldsymbol{\beta}}, \tilde{\gamma}, \tilde{\boldsymbol{\zeta}})$ denote the score vectors and Jacobian matrix evaluated at the solutions to the ARGL estimating equations. A cluster-robust sandwich estimator is then given by:

$$\mathbf{V}^{ARGL} = \tilde{\mathbf{J}}^{-1} \left(\sum_{j=1}^{J} \tilde{\mathbf{M}}_j \tilde{\mathbf{M}}_j' \right) \left(\tilde{\mathbf{J}}^{-1} \right)'. \tag{B5}$$

The Jacobian of (14) is given by

$$\mathbf{J} = \sum_{j=1}^{J} \begin{bmatrix} \mathbf{J}_j^{\beta\beta'} & \mathbf{J}_j^{\beta\gamma'} & \mathbf{J}_j^{\beta\zeta'} \\ \mathbf{J}_j^{\gamma\beta'} & \mathbf{J}_j^{\gamma\gamma'} & \mathbf{J}_j^{\gamma\zeta'} \\ \mathbf{J}_j^{\zeta\beta'} & \mathbf{J}_j^{\zeta\gamma'} & \mathbf{J}_j^{\zeta\zeta'}, \end{bmatrix}$$

where

$$\begin{aligned}
\mathbf{J}_j^{\beta\beta'} &= -\sum_{i=1}^{k_j} a_{ij} \frac{\mathbf{x}_{ij}' \mathbf{x}_{ij}}{w_{ij}\eta_{ij}} \\
\mathbf{J}_j^{\beta\gamma} &= -\sum_{i=1}^{k_j} a_{ij} \mathbf{x}_{ij}' \tau^2 \left(\frac{y_{ij} - \mu_{ij}}{w_{ij}\eta_{ij}^2} \right) \\
\mathbf{J}_j^{\beta\zeta_h} &= -\sum_{i=1}^{k_j} a_{ij} \mathbf{x}_{ij}' \lambda_h \left(\frac{y_{ij} - \mu_{ij}}{w_{ij}^2 \eta_{ij}} \right) \\
\mathbf{J}_j^{\gamma\beta'} &= \left(\mathbf{J}_j^{\beta\gamma} \right)' \\
\mathbf{J}_j^{\gamma\gamma} &= -\sum_{i=1}^{k_j} a_{ij} \left[\frac{\tau^4}{w_{ij}} \left(\frac{2(y_{ij} - \mu_{ij})^2 - \eta_{ij}}{2\eta_{ij}^3} \right) - \frac{\tau^2}{w_{ij}} \left(\frac{(y_{ij} - \mu_{ij})^2 - \eta}{2\eta_{ij}^2} \right) \right] \\
\mathbf{J}_j^{\gamma\zeta_h} &= -\sum_{i=1}^{k_j} a_{ij} \frac{\tau^2 \lambda_h}{w_{ij}^2} \left(\frac{(y_{ij} - \mu_{ij})^2 - \eta_{ij}}{2\eta_{ij}^2} \right) \\
\mathbf{J}_j^{\zeta_h\beta'} &= \left(\mathbf{H}_j^{\beta\zeta_h} \right)' \\
\mathbf{J}_j^{\zeta_h\gamma} &= \mathbf{H}_j^{\gamma\zeta_h} \\
\mathbf{J}_j^{\zeta_f\zeta_h} &= \mathbf{H}_j^{\zeta_f\zeta_h'},
\end{aligned}$$

with $\mathbf{H}_j^{\beta\zeta_h}$, $\mathbf{H}_j^{\gamma\zeta_h}$, and $\mathbf{H}_j^{\zeta_f\zeta_h}$ as given in Equations (A13), (A15), and (A16).

C Further details about bootstrapping

C.1 Other bootstrap sampling methods

The main manuscript described a two-stage cluster bootstrapping technique. We also considered two other bootstrap sampling schemes.

In the clustered bootstrap scheme, each pseudo-sample is generated by randomly drawing J clusters of observations with replacement from the original sample. In contrast to the two-stage cluster bootstrap, sampled clusters are left intact rather than perturbed. Because clusters are drawn with replacement, some will necessarily be included multiple times and some clusters might not be included in a given pseudo-sample. Following the notation in the main text, let $a_j^{(b)}$ be a first-stage weight for cluster j and $a_{ij}^{(b)}$ be the weight assigned to observation i in cluster j for pseudo-sample b. The clustered bootstrap process is then equivalent to drawing $a_1^{(b)}, ..., a_J^{(b)}$ from a multinomial distribution with J trials and equal probability on each of J categories and then setting $a_{ij}^{(b)} = a_j^{(b)}$ for $i = 1, ..., k_j$ and $j = 1, ..., J$.

The fractional random weight bootstrap follows a very similar process in which each pseudo-sample is generated by assigning a random weight to every cluster. Rather than using a multinomial distribution, the weights follow independent exponential distributions with mean 1, so that the weights for pseudo-sample b are generated as $a_j^{(b)} \sim Exp(1)$, with $a_{ij}^{(b)} = a_j^{(b)}$ for $i = 1, ..., k_j$ and $j = 1, ..., J$. A crucial difference between these bootstrap techniques is that the fractional random weight bootstrap puts strictly positive weight on every cluster of observations in every pseudo-sample, whereas the clustered bootstrap and two-stage bootstrap can assign zero weight to some clusters.[1] If the existence of an estimator hinges on the inclusion of one or a small number of clusters, this will create computational problems for the non-parametric bootstrap but not for the fractional random weight bootstrap.

C.2 Bootstrap confidence interval construction

We describe three methods for constructing a $1-2\alpha$ confidence interval (CI) from a set of B bootstrap replications. Consider a parameter θ that is a scalar component of $\boldsymbol{\beta}$, γ, or $\boldsymbol{\zeta}$. Let $\hat{\theta}$ denote an estimator of θ with sandwich variance estimator V. Let $\hat{\theta}^*_b$ denote the same estimator computed from bootstrap pseudo-sample b, with corresponding sandwich variance estimator V^*_b. Let $\hat{\theta}^*_{(1)}, ..., \hat{\theta}^*_{(B)}$ denote the pseudo-sample estimators sorted in ascending order. An estimator for the standard error of $\hat{\theta}$ can be computed by taking the standard deviation of the $\hat{\theta}^*_1, ..., \hat{\theta}^*_B$.

First, the percentile CI is calculated by taking the α and $1-\alpha$ quantiles of the bootstrap distribution, with end-points

$$\left[\hat{\theta}^*_{((B+1)\alpha)}, \hat{\theta}^*_{((B+1)(1-\alpha))}\right].$$

Second, the so-called "basic" CI pivots the bootstrap distribution around the original estimator $\hat{\theta}$. Its end-points are given by

$$\left[2\hat{\theta} - \hat{\theta}^*_{((B+1)(1-\alpha)}, 2\hat{\theta} - \hat{\theta}^*_{((B+1)\alpha)}\right].$$

Third, a studentized CI uses the bootstrap distribution of t-statistics rather than point estimators. The t statistic for pseudo-sample b is computed as $t^*_b = \left(\hat{\theta}^*_b - \hat{\theta}\right)/\sqrt{V^*_b}$. The studentized CI is computed using the percentiles of the bootstrap distribution of $t^*_1, ..., t^*_B$, taking

$$\left[\hat{\theta} - \sqrt{V} \times t^*_{((B+1)(1-\alpha)}, \ \hat{\theta} - \sqrt{V} \times t^*_{((B+1)\alpha)}\right].$$

Fourth, the bias-corrected-and-accelerated (BCa) CI is similar to the percentile CI in that its end-points are defined by quantiles of the bootstrap distribution. However, instead using the α and $1-\alpha$ quantiles, it uses quantiles that are adjusted to take into account the bias of the estimator and the degree to which its sampling variance is related to the underlying parameter, as measured using an acceleration coefficient. These adjustments are defined in terms of the empirical influence function, which we approximate using a leave-one-cluster-out jackknife. The jackknife influence value for cluster j is $\hat{\theta} - \hat{\theta}^+_{-j}$, where $\hat{\theta}^+_{-j}$ denotes the estimator of θ computed while leaving out the observations in cluster j for $j = 1, ..., J$. The acceleration coefficient is then

$$\hat{a} = \frac{\sum_{j=1}^{J} \left(\hat{\theta} - \hat{\theta}^+_{-j}\right)^3}{6\left[\sum_{j=1}^{J} \left(\hat{\theta} - \hat{\theta}^+_{-j}\right)^2\right]^{3/2}}.$$

The bias coefficient is calculated as the proportion of the bootstrap distribution that falls below the original estimator:

$$\hat{\beta} = \frac{1}{B}\sum_{b=1}^{B} I\left(\hat{\theta}^*_b < \hat{\theta}\right).$$

With the acceleration and bias coefficients defined, define the adjustment function $f(\alpha)$ as

$$f(\alpha) = \Phi\left(\Phi^{-1}(\hat{\beta}) + \frac{\Phi^{-1}(\hat{\beta}) + \Phi^{-1}(\alpha)}{1 - \hat{a}\left[\Phi^{-1}(\hat{\beta}) + \Phi^{-1}(\alpha)\right]}\right)$$

for $0 < \alpha < 1$. The end-points of the BCa CI are then given by

$$\left[\hat{\theta}^*_{((B+1)\times f(\alpha))}, \hat{\theta}^*_{((B+1)\times f(1-\alpha))}\right].$$

Notably, the basic and studentized confidence intervals depend on the scale of the parameter θ, and the accuracy of their coverage levels therefore depends on the parameterization. In contrast, the percentile and bias-corrected-and-accelerated confidence intervals are invariant to transformation of θ.

C.3 Extrapolating coverage rates

The simulation results reported in the main manuscript used an extrapolation technique suggested by Boos and Zhang[2] to estimate the coverage levels of bootstrap confidence intervals for $B = 1999$ bootstrap re-samples, when each replication of the simulation involved only $B = 299$ re-samples. We used $R = 2400$ replications of the simulation process. For each replication, we computed an estimator $\hat{\theta}$ of parameter θ and a set of bootstrap re-samples $\hat{\theta}_b^*$ for $b = 1, ..., 299$. With these data, we can construct a confidence interval for θ by drawing $d \leq B$ of the bootstrap re-samples at random without replacement, then using these replications in the calculations described in Section C.2. We repeat this process for $d = 49, 99, 199, 299$. Let C^{dr} be an indicator for whether the confidence interval constructed from d re-samples covers θ, for $r = 1, ..., R$. To extrapolate coverage, we fit a linear regression

$$C_{dr} = \phi_0 + \phi_1 \left(\frac{1}{d} - \frac{1}{1999}\right) + e_{dr}$$

by ordinary least squares, with standard errors clustered by replication r to account for dependence in C_{dr} computed from the same replication of the simulation process. We take the estimate of ϕ_0 as the coverage rate of confidence intervals constructed from $B = 1999$ re-samples.

To validate this extrapolation technique, we ran additional simulations where we generated confidence intervals based on $B = 1999$ re-samples and computed coverage rates directly. Due to high computational demand, we conducted this exercise for a small subset of the conditions examined in the full simulation study, which we selected because they varied in the estimate coverage rates of percentile confidence intervals. Specifically, we looked at conditions with the univariate selection process ($\psi = 0$), $J = 15$ studies, an average effect size of $\mu = 0.2$, typical primary study sample sizes, a heterogeneity ratio of $\omega^2/\tau_B^2 = 0.5$, outcome correlation $\rho = 0.8$, between-study heterogeneity of $\tau_B \in \{0.05, 0.45\}$, and probability of selection $\lambda_1 \in \{0.05, 0.20\}$. We used the two-stage bootstrap re-sampling process because it showed the best performance in the larger simulation study.

Figures C1 and C2 illustrate the extrapolation results for the penalized maximum

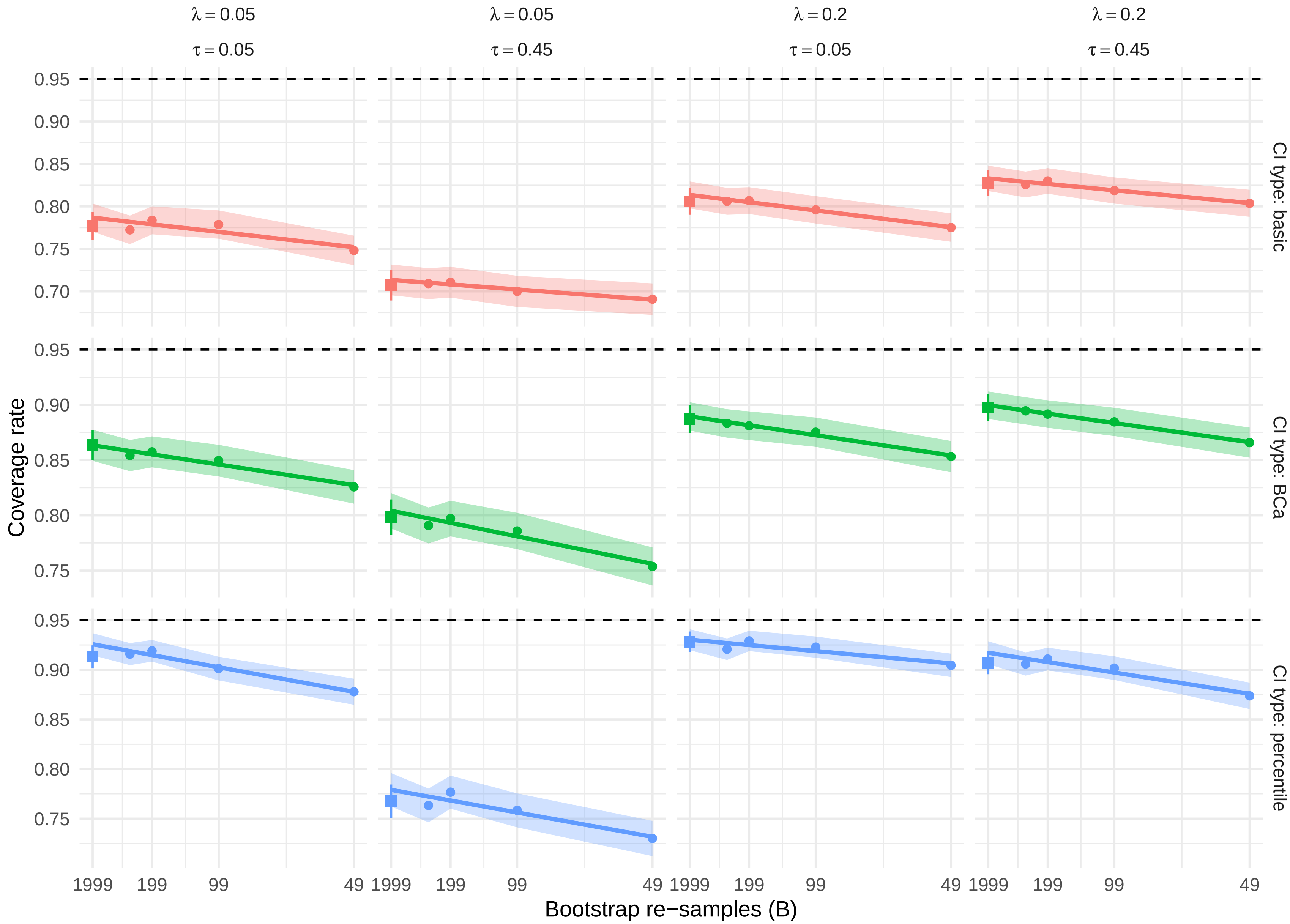

Figure C1

Actual and extrapolated coverage rates of selected bootstrap confidence intervals based on composite maximum likelihood estimation, using two-stage bootstrap resampling. Circular dots represent actual coverage rates used to extrapolate coverage to $B = 1999$ *re-samples, with colored bands indicating point-wise Monte Carlo intervals. Squares represent actual coverage rates using* $B = 1999$ *re-samples, with whiskers indicating point-wise Monte carlo intervals, based on 2400 replications.*

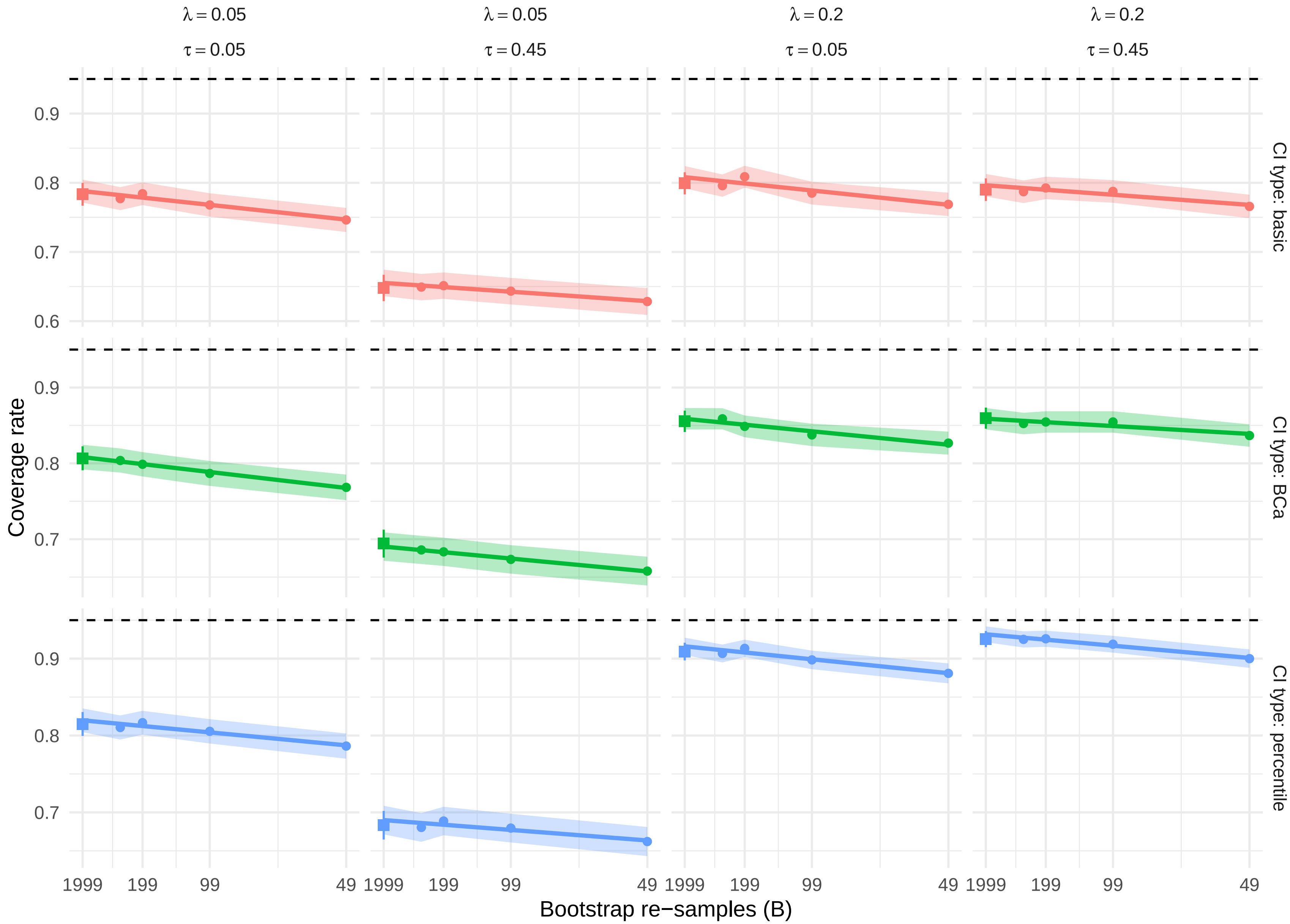

Figure C2

Actual and extrapolated coverage rates of selected bootstrap confidence intervals based on augmented reweighted Gaussian likelihood estimation, using two-stage bootstrap resampling. Circular dots represent actual coverage rates used to extrapolate coverage to $B = 1999$ *re-samples, with colored bands indicating point-wise Monte Carlo intervals. Squares represent actual coverage rates using* $B = 1999$ *re-samples, with whiskers indicating point-wise Monte carlo intervals, based on 2400 replications.*

likelihood and augmented, reweighted Gaussian likelihood estimators, respectively. In each figure, the circular points show the estimated coverage rates of a specific type of bootstrap confidence interval computed from $d = 49, 99, 199$, or 299 bootstrap re-samples, with bands representing point-wise Monte Carlo intervals; these data are drawn from the larger simulation study and use $R = 2400$ replications. The horizontal axis of each panel is scaled in proportion to $1/d$, and the straight line in each panel depicts the linear regression fit used for extrapolation. On the left edge of each panel, a square point and vertical whiskers indicate the coverage rates computed from the additional simulations with $B = 1999$ re-samples. Comparing the square points to the linear extrapolations provides a sense of the accuracy of the extrapolation technique.

It can be seen that nearly all coverage rates from the additional simulations fall within the Monte Carlo intervals of the extrapolation. Across conditions, estimators, and confidence interval types, discrepancies between extrapolated coverage and directly estimated coverage ranged from -0.4 to 1.2 percentage points, with an average of 0.5 and root mean-squared error of 0.7 percentage points. Thus, the extrapolations computed in the large simulation study appear to accurately predict the coverage rates computed from the additional simulations with $B = 1999$ re-samples. Furthermore, the size of the extrapolations is generally quite small. For the percentile bootstrap CIs, the difference between the extrapolated coverage rate and the measured coverage rate for $B = 299$ bootstrap replications is no more than 1.6 percentage points for the penalized maximum likelihood estimator and no more than 1.0 percentage points for the augmented, reweighted Gaussian likelihood estimator.

D Additional simulation results for estimators of average effect size (μ)

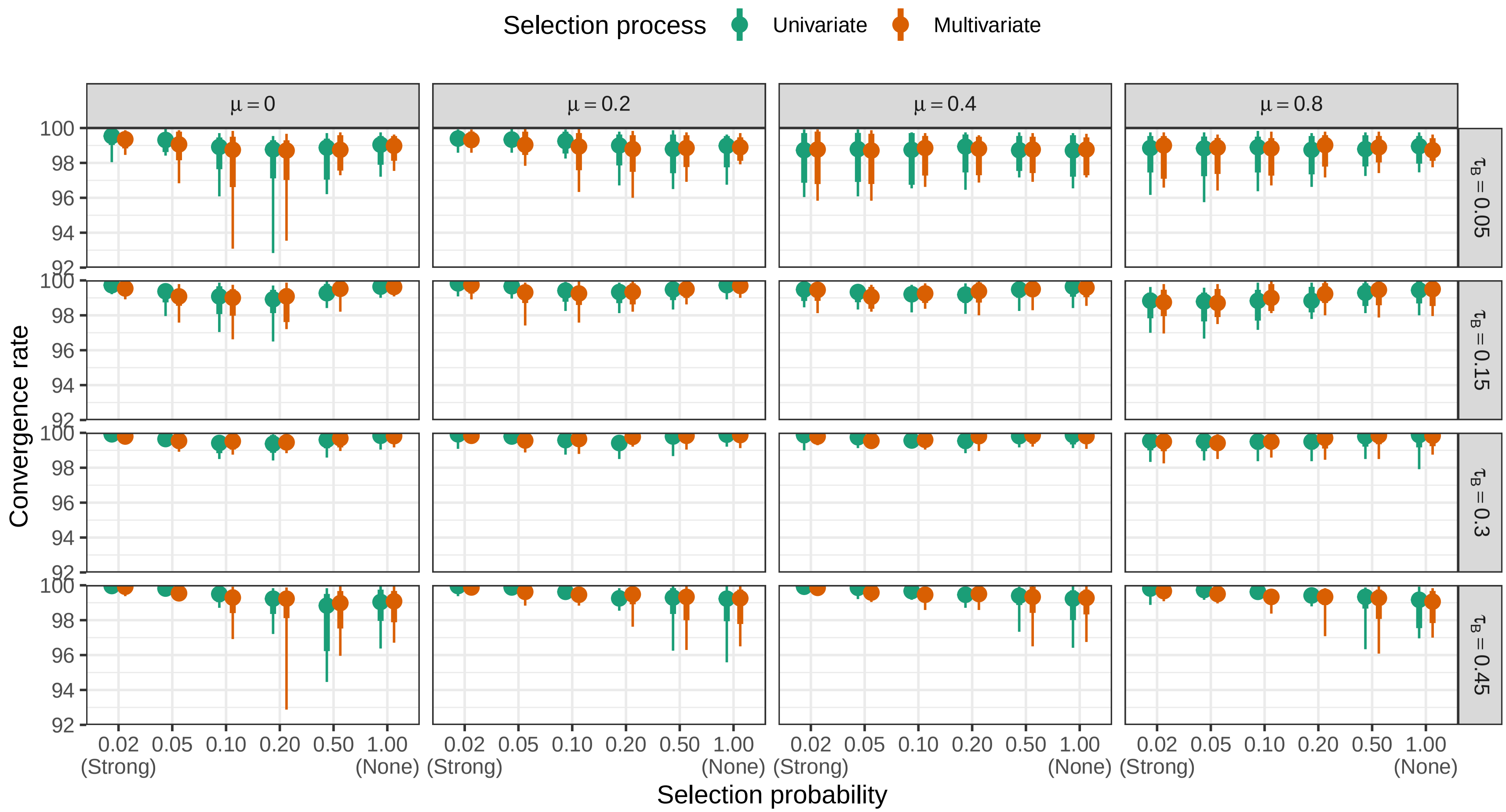

Figure D1

Convergence rates of PML estimators by selection probability, average SMD, and between-study heterogeneity. Points correspond to median convergence rates; thin lines correspond to range of convergence rates; thick lines correspond to inter-decile range.

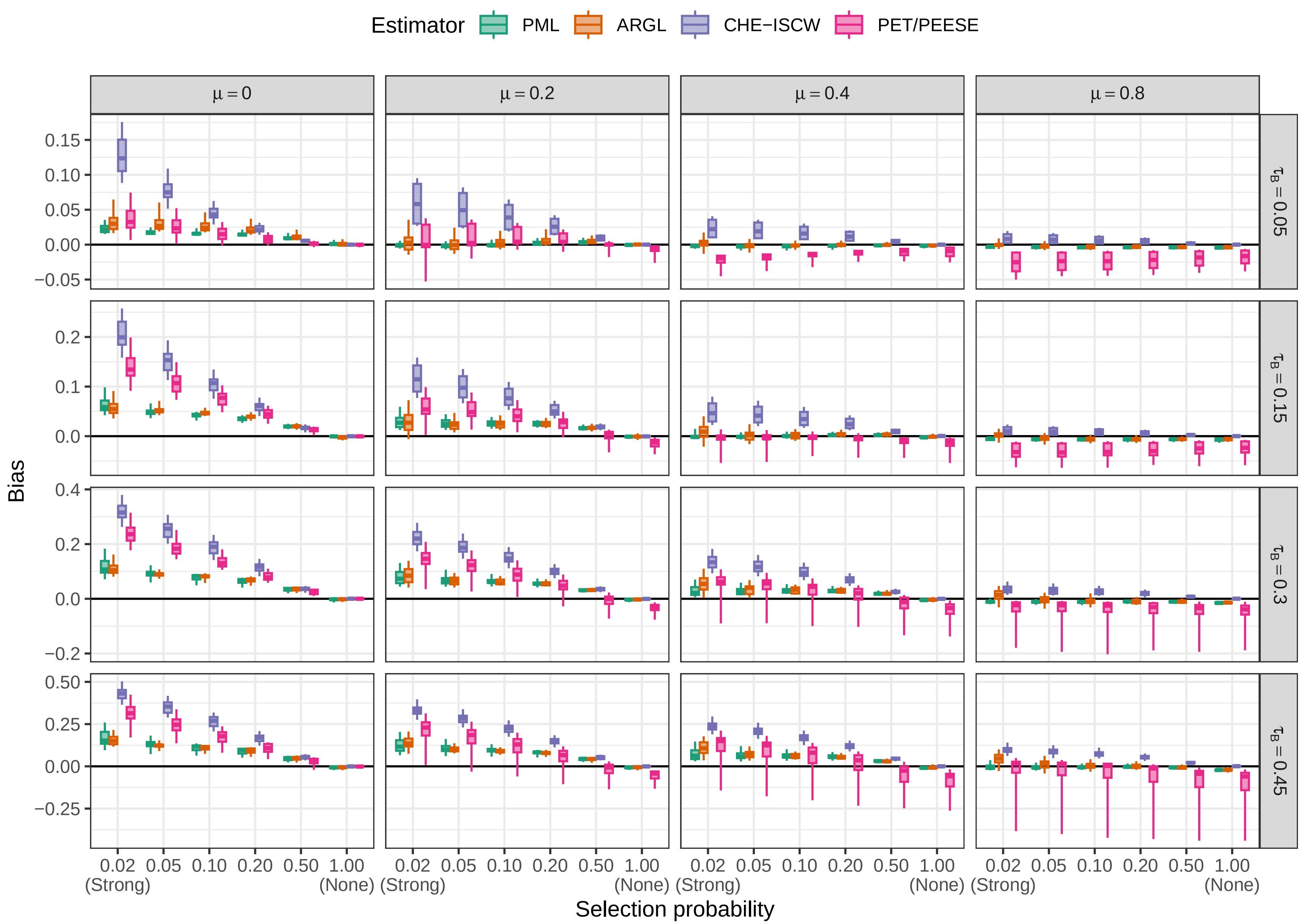

Figure D2

Bias for estimators of average effect size by selection probability, average SMD, and between-study heterogeneity across scenarios with a multivariate selection process ($\psi = 1$)

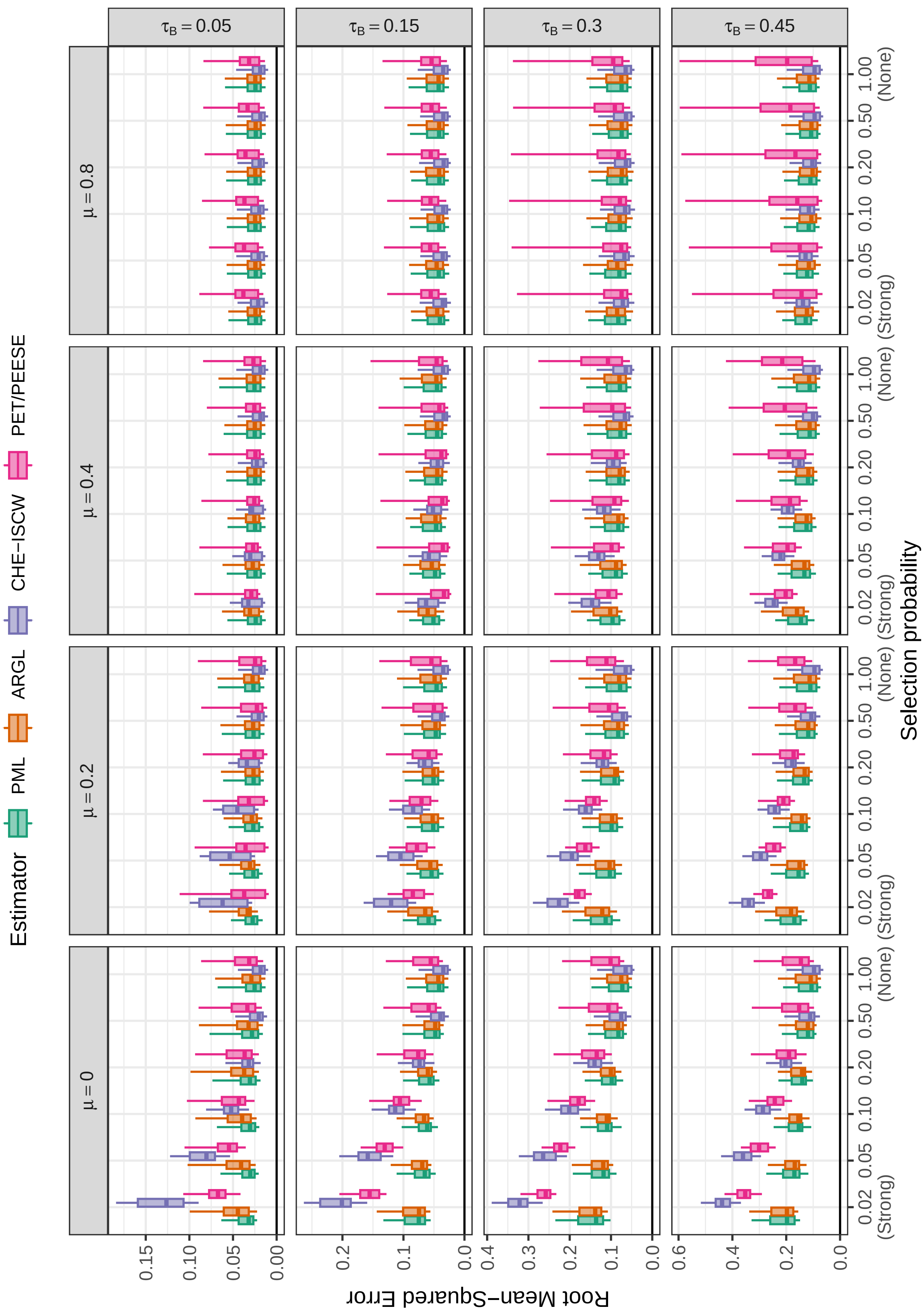

Figure D3

Root mean-squared error for estimators of average effect size by selection probability, average SMD, and between-study heterogeneity across scenarios with a multivariate selection process ($\psi = 1$)

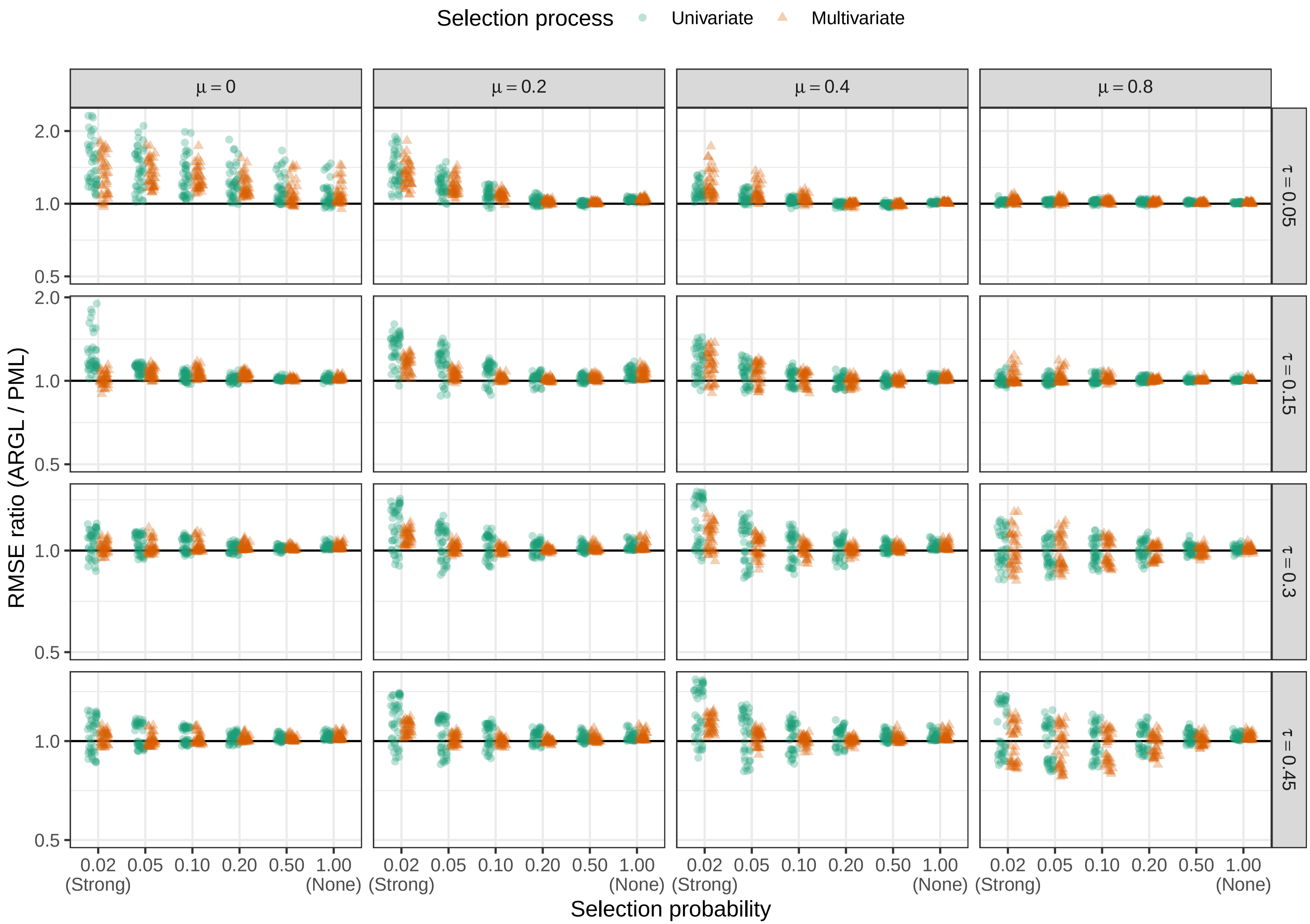

Figure D4

Ratio of root mean-squared error for ARGL estimator to root mean-squared error of PML estimator by selection process, selection probability, number of studies, average SMD, and between-study heterogeneity

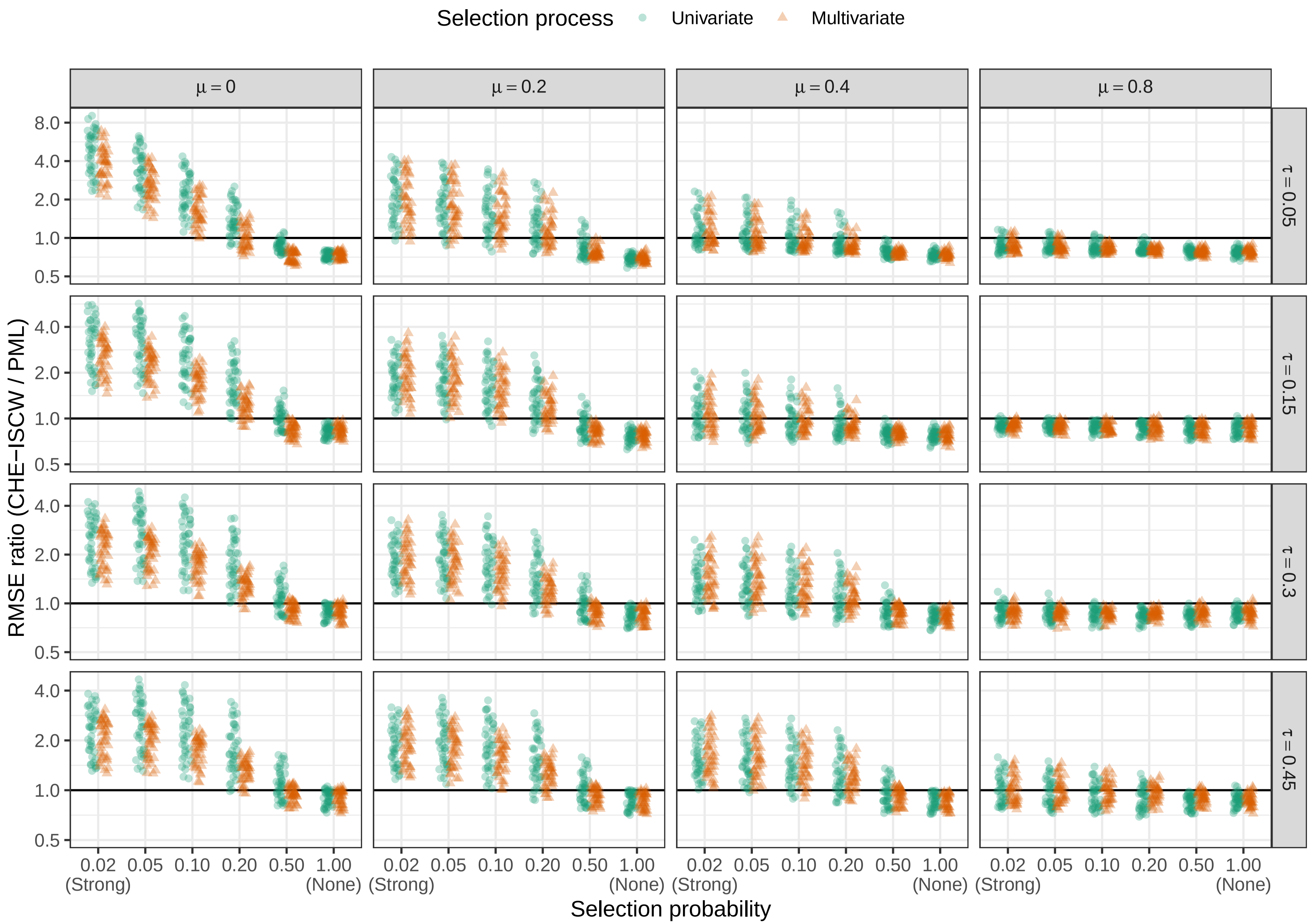

Figure D5

Ratio of root mean-squared error for CHE-ISCW estimator to root mean-squared error of PML estimator by selection process, selection probability, number of studies, average SMD, and between-study heterogeneity

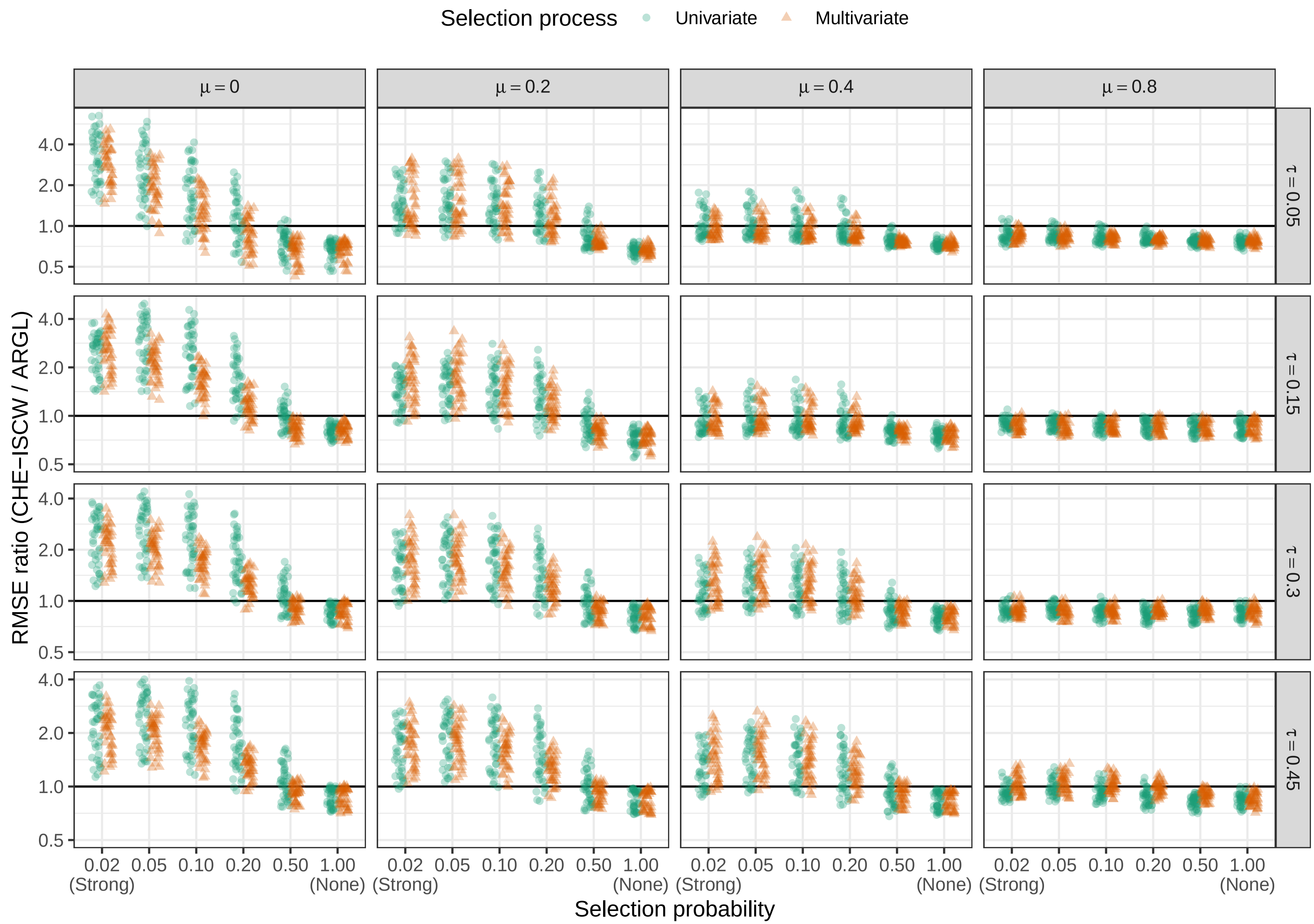

Figure D6

Ratio of root mean-squared error for CHE-ISCW estimator to root mean-squared error of ARGL estimator by selection process, selection probability, number of studies, average SMD, and between-study heterogeneity

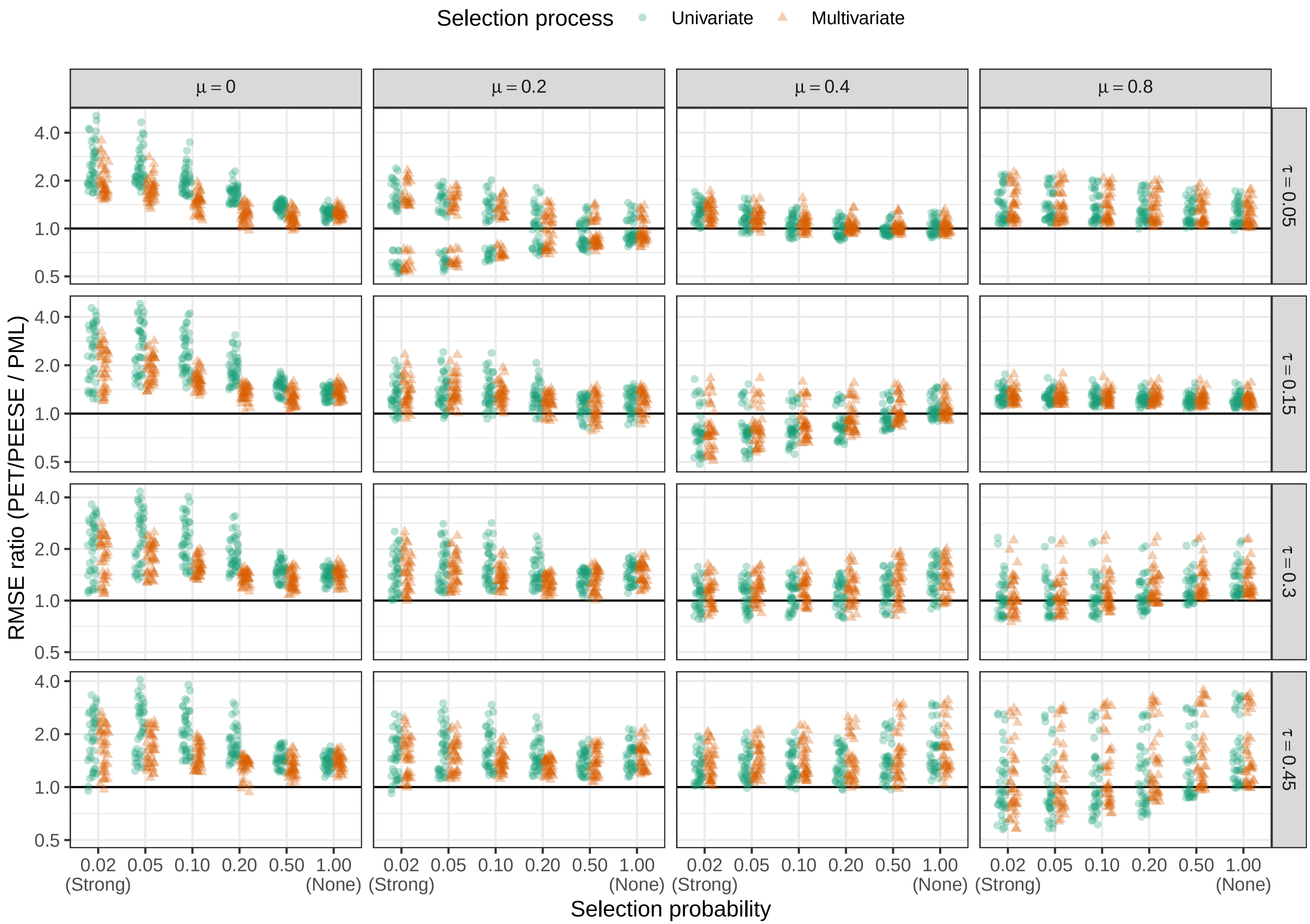

Figure D7

Ratio of root mean-squared error for PET/PEESE estimator to root mean-squared error of PML estimator by selection process, selection probability, number of studies, average SMD, and between-study heterogeneity

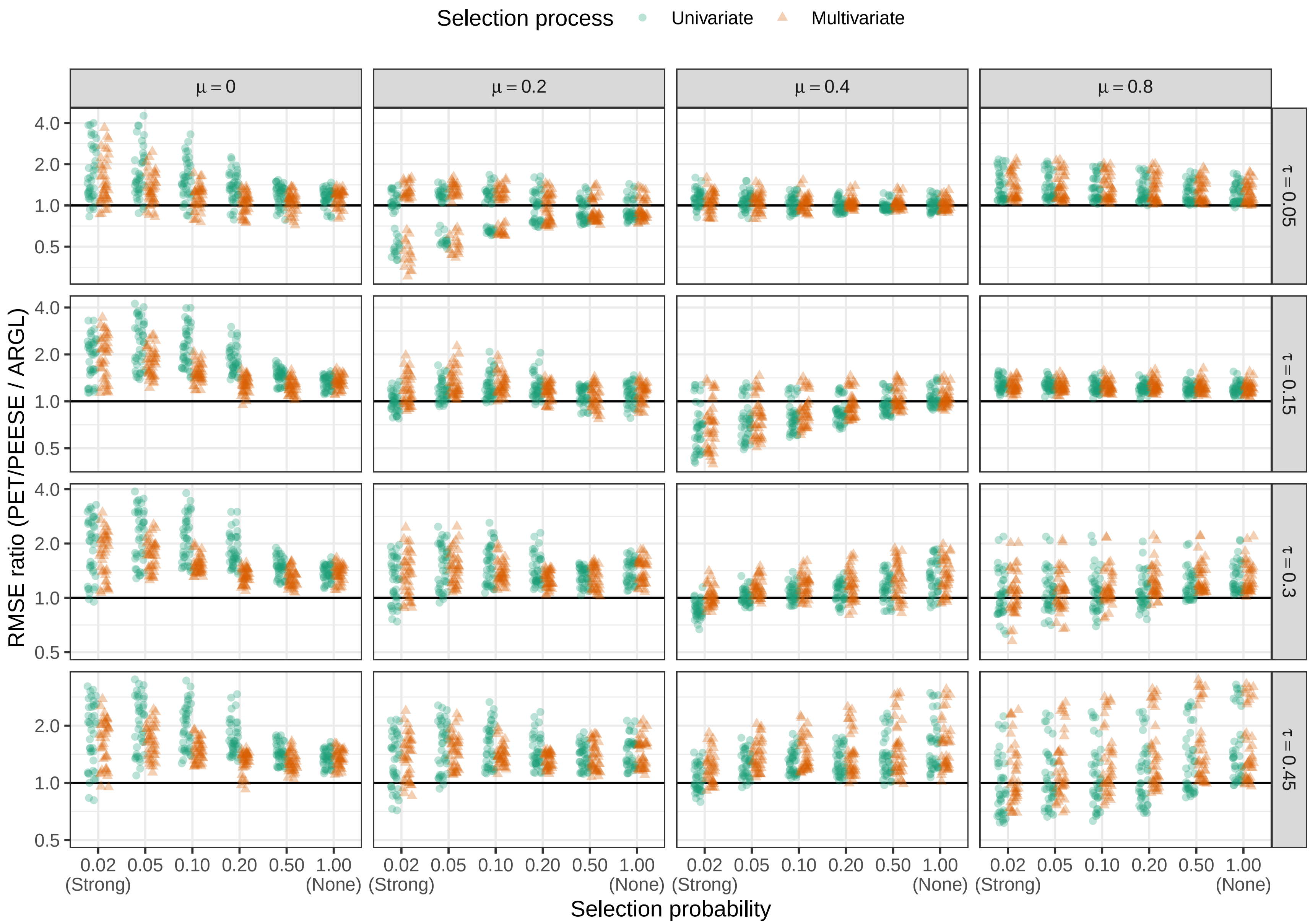

Figure D8

Ratio of root mean-squared error for PET/PEESE estimator to root mean-squared error of ARGL estimator by selection process, selection probability, number of studies, average SMD, and between-study heterogeneity

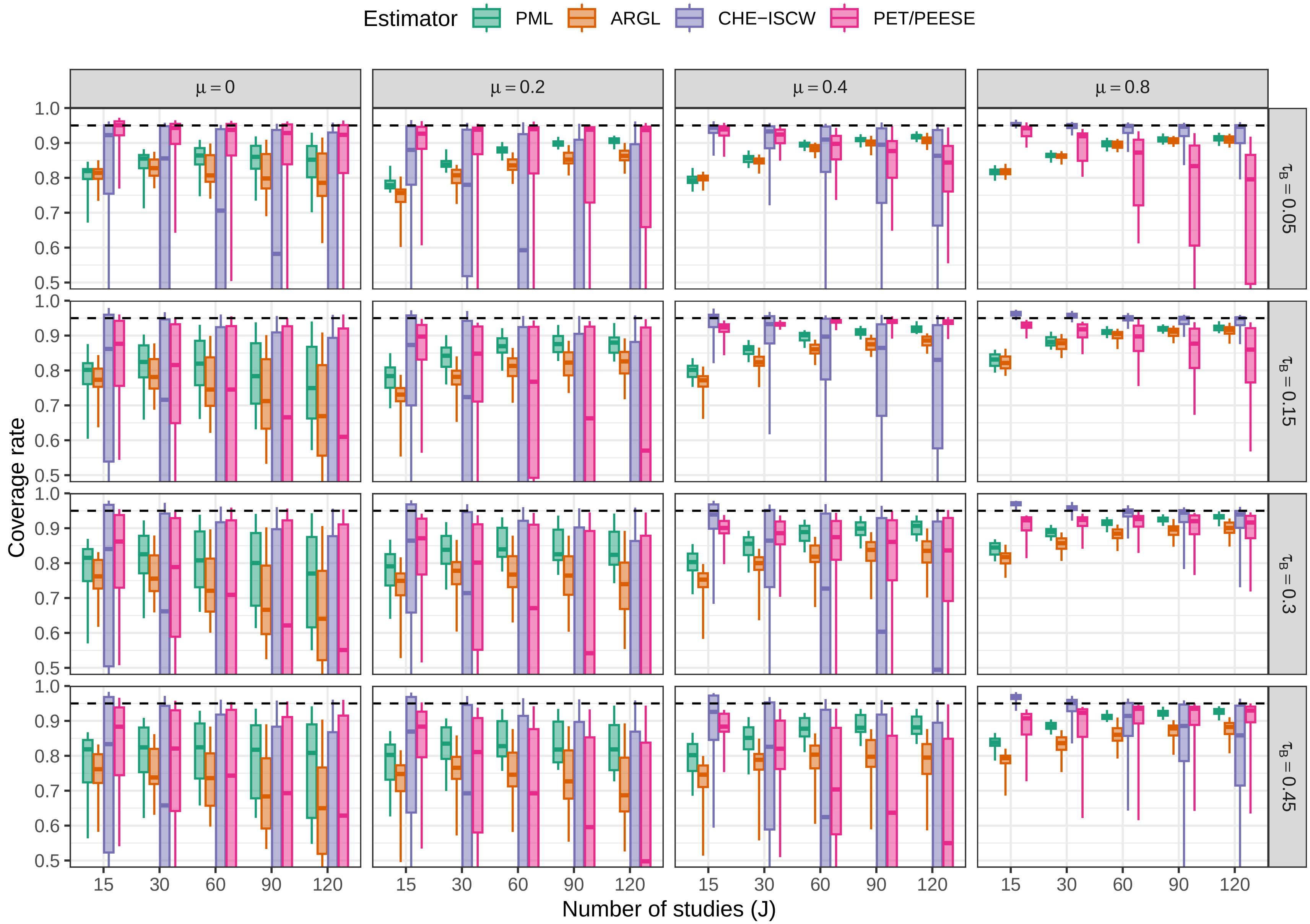

Figure D9

Coverage levels of confidence intervals based for average effect size based on cluster-robust variance approximations, by number of studies, average SMD, and between-study heterogeneity across scenarios with a multivariate selection process ($\psi = 1$). Dashed lines correspond to the nominal confidence level of 0.95. Coverage rates of the CHE-ISCW and PET/PEESE intervals are not depicted when they fall below 0.5

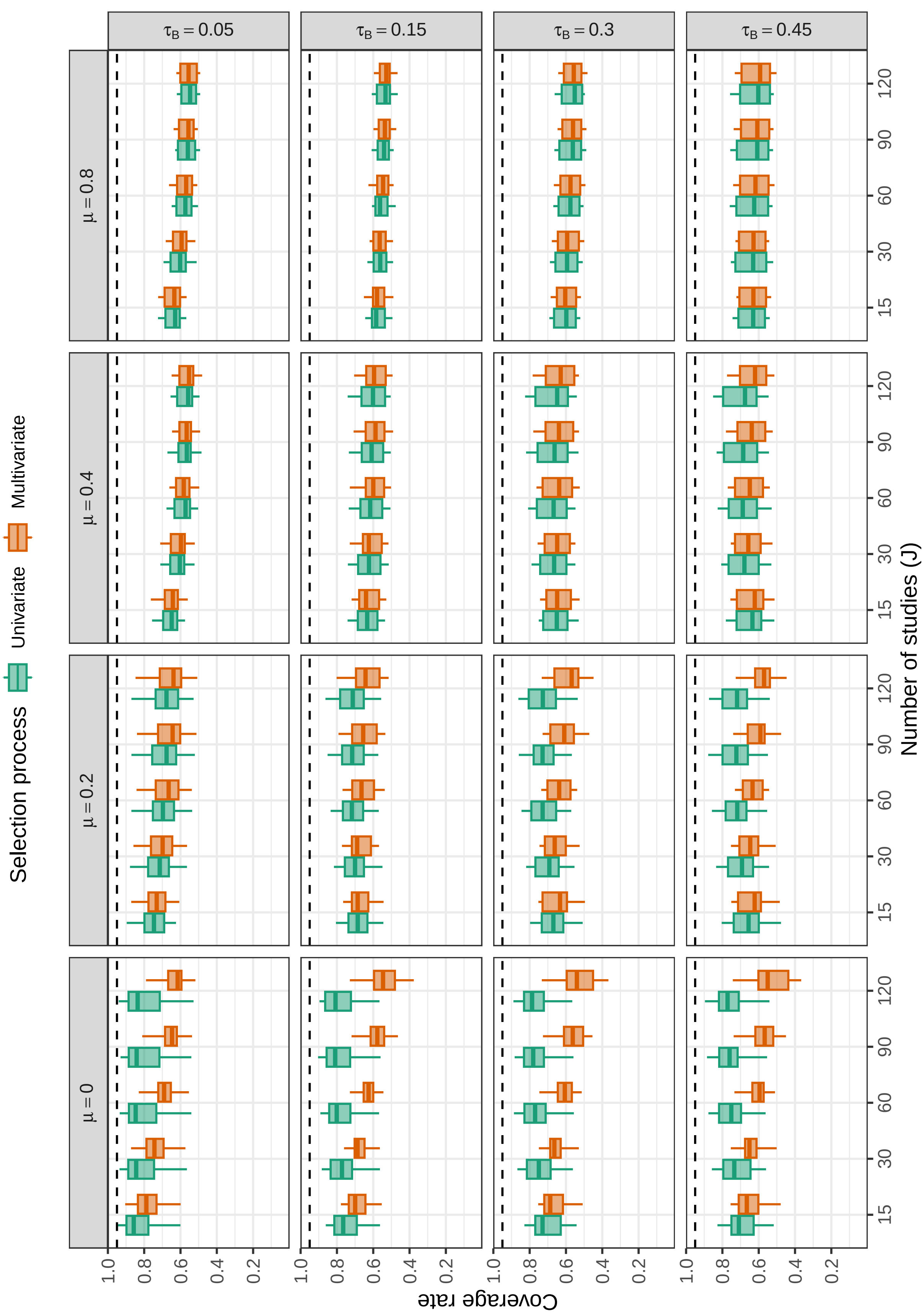

Figure D10

Coverage levels of confidence intervals for average effect size based on model-based variance estimation, by number of studies, average SMD, and between-study heterogeneity under a univariate or multivariate selection process. Dashed lines correspond to the nominal confidence level of 0.95.

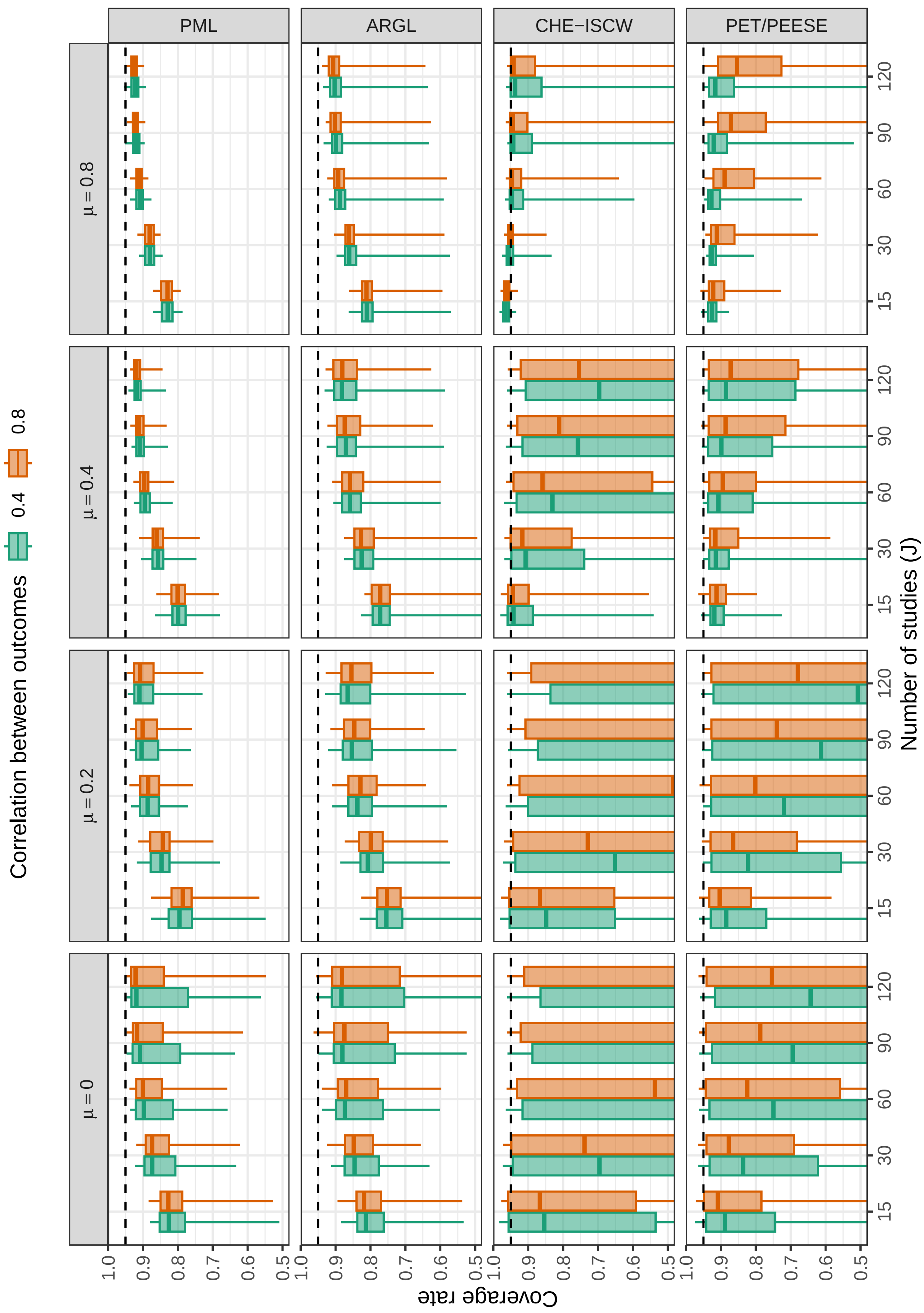

Figure D11

Coverage levels of confidence intervals based for average effect size based on cluster-robust variance approximations, by number of studies, average SMD, and average correlation between outcomes. Dashed lines correspond to the nominal confidence level of 0.95.

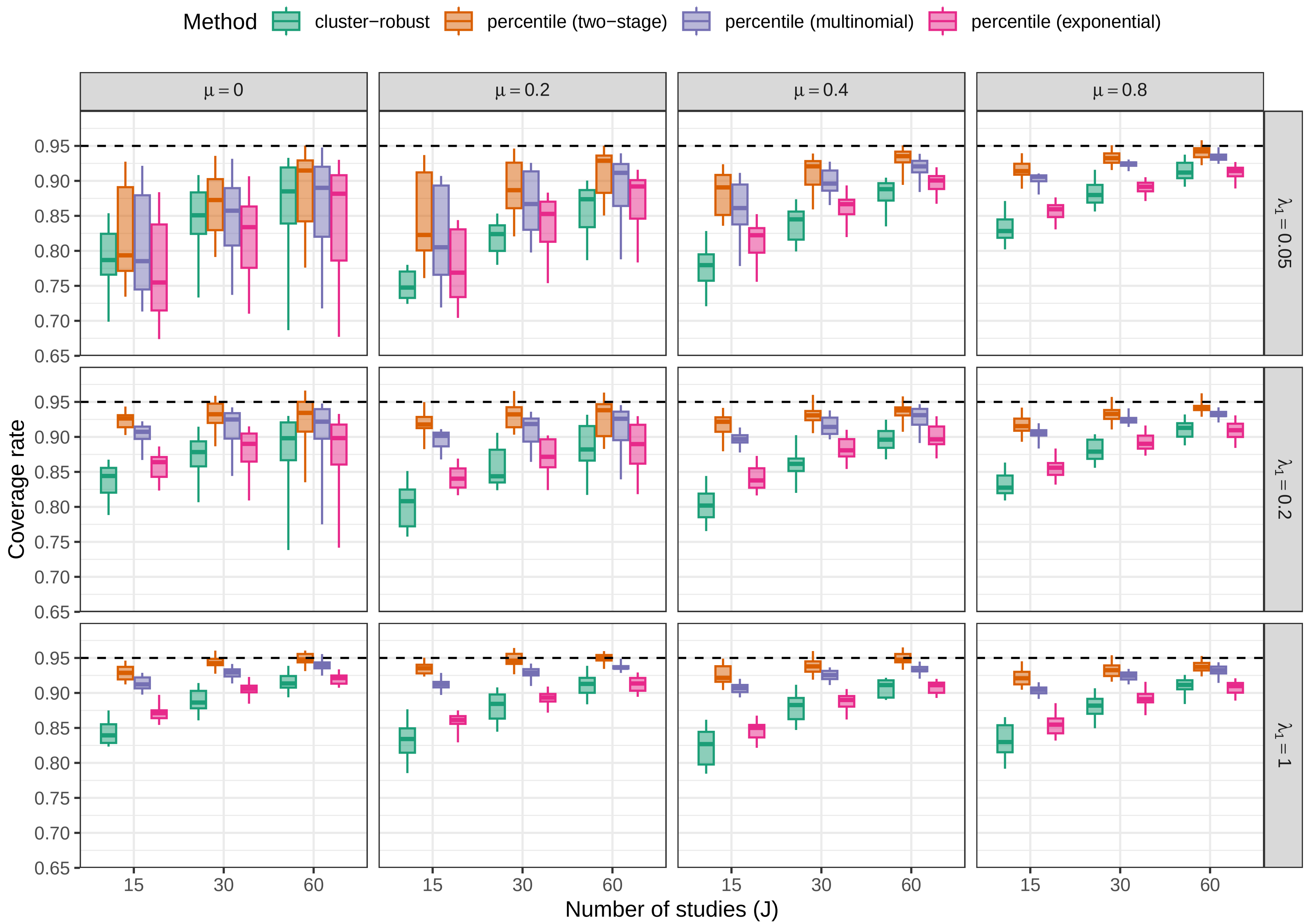

Figure D12

Coverage levels of confidence intervals based on the PML estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a univariate selection process ($\psi = 0$). Dashed lines correspond to the nominal confidence level of 0.95.

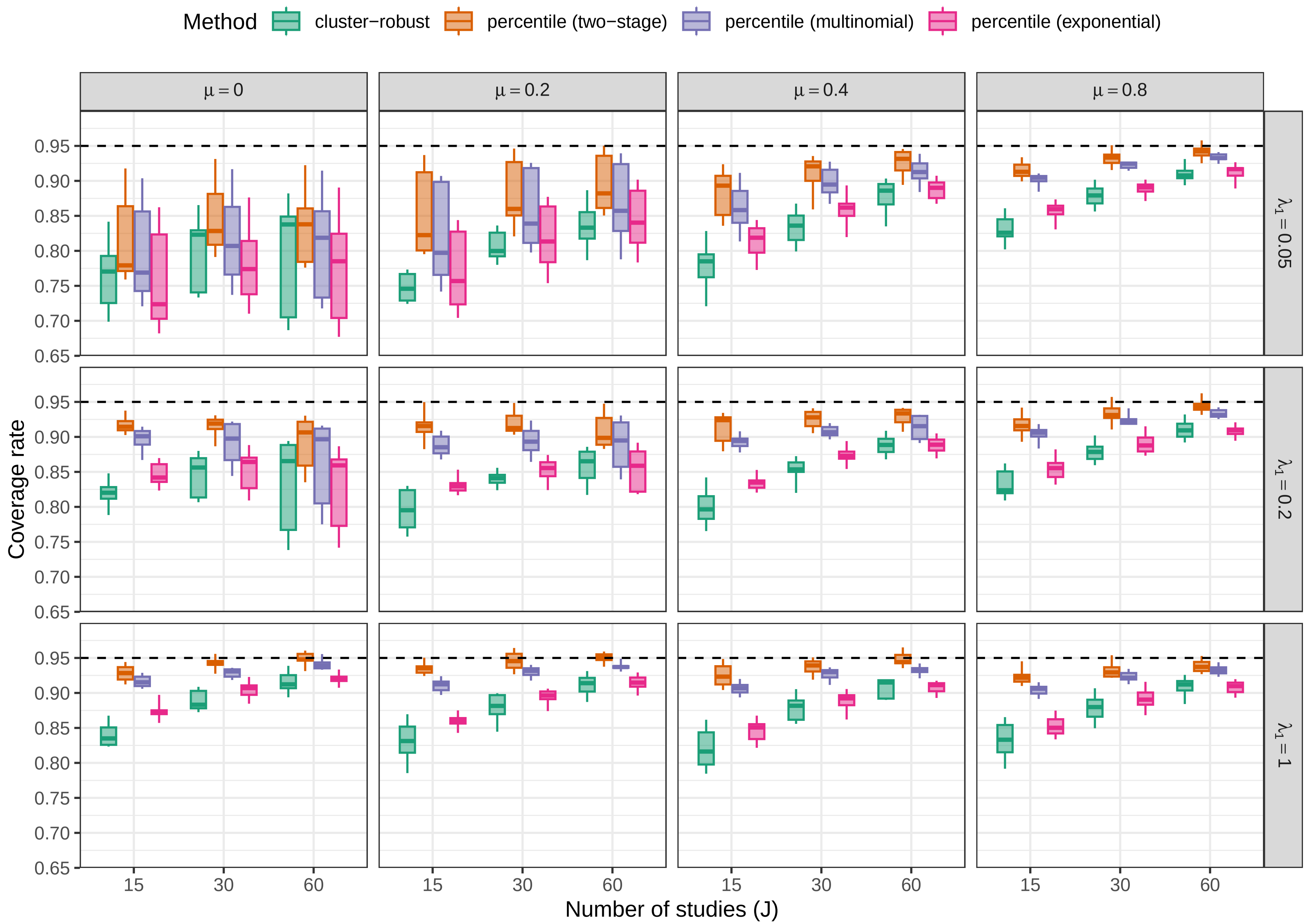

Figure D13

Coverage levels of confidence intervals based on the PML estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a multivariate selection process ($\psi = 1$). Dashed lines correspond to the nominal confidence level of 0.95.

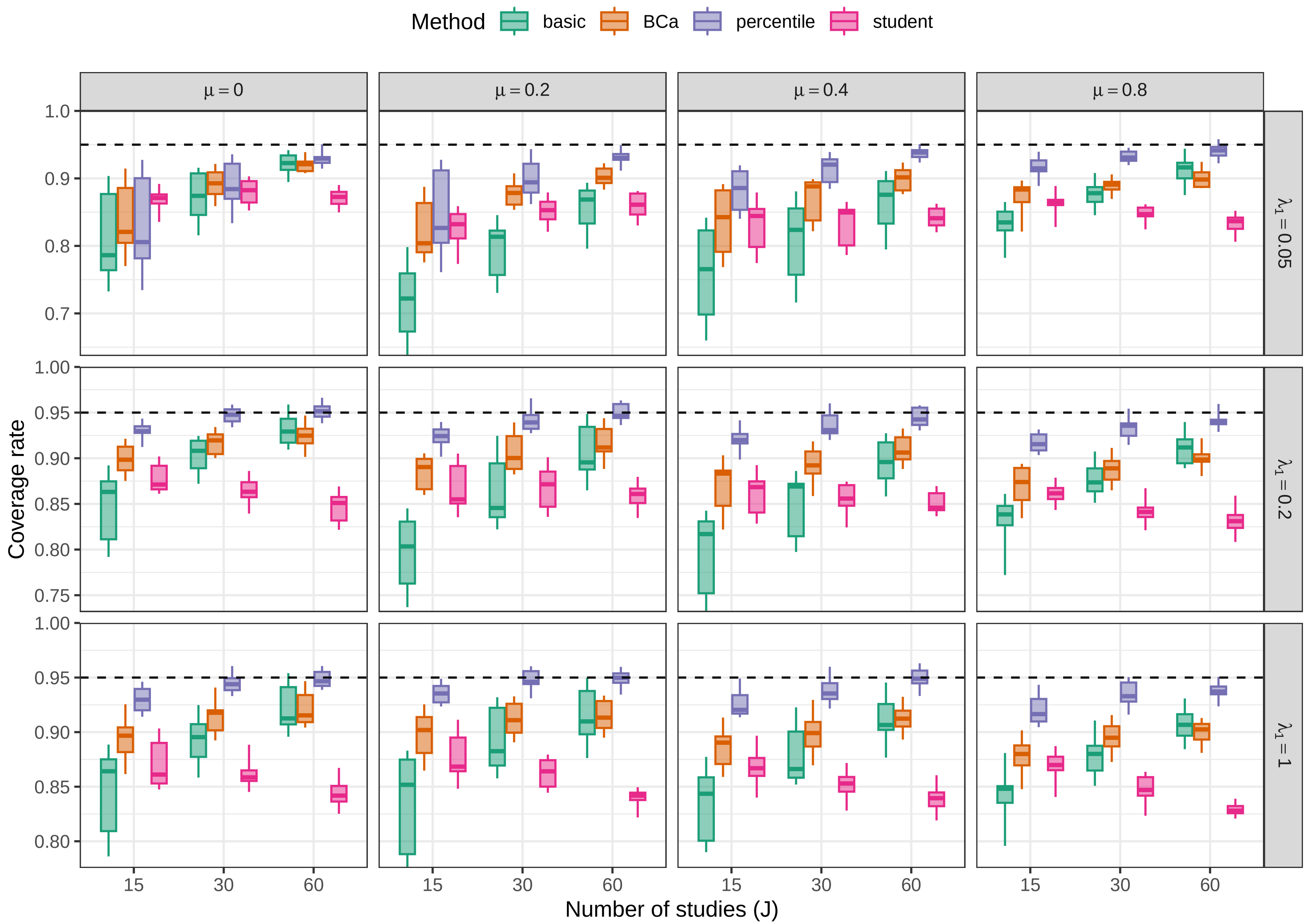

Figure D14

Coverage levels of two-stage bootstrap confidence intervals based on the PML estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a univariate selection process ($\psi = 0$). Dashed lines correspond to the nominal confidence level of 0.95.

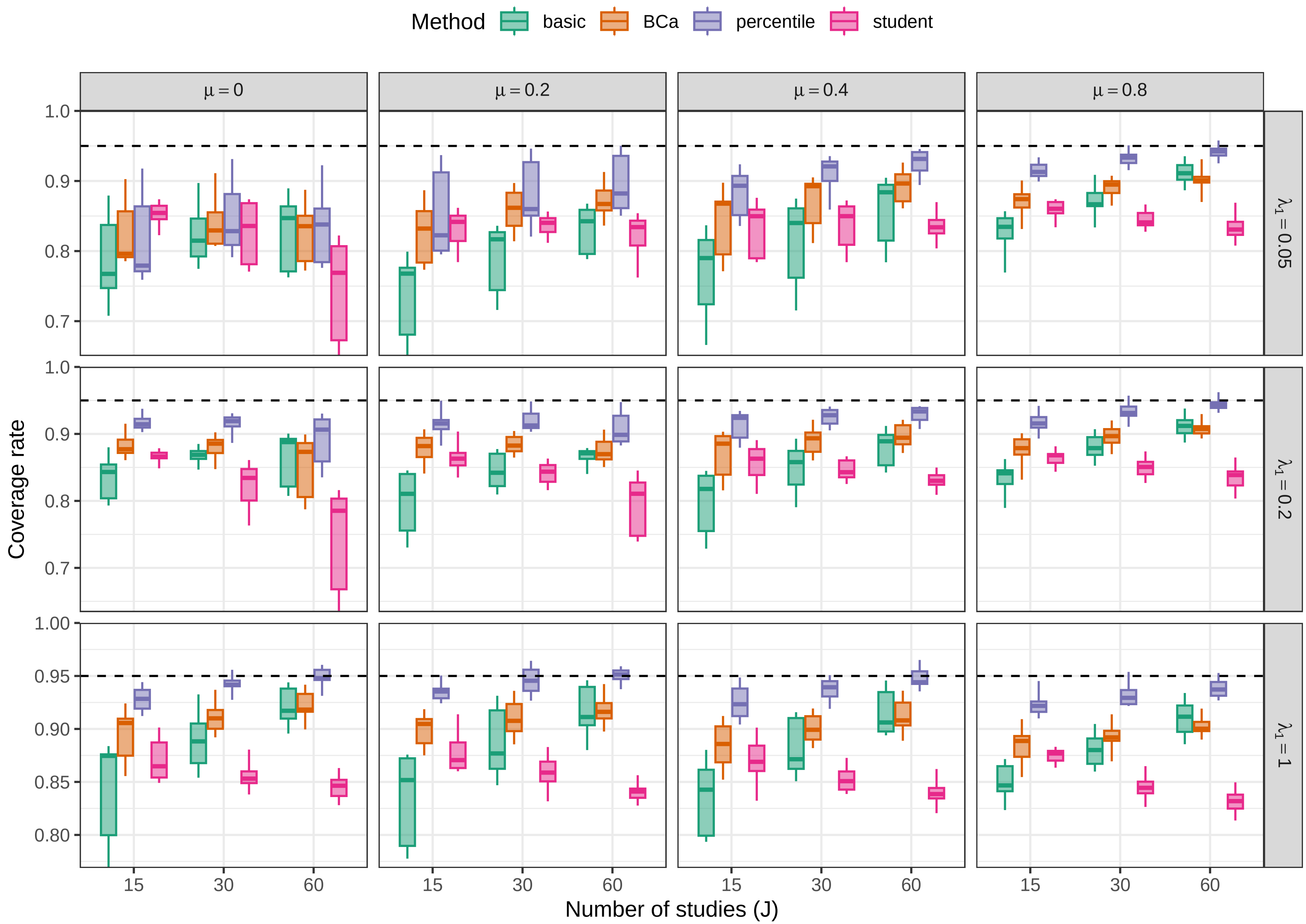

Figure D15

Coverage levels of two-stage bootstrap confidence intervals based on the PML estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a multivariate selection process ($\psi = 1$). Dashed lines correspond to the nominal confidence level of 0.95.

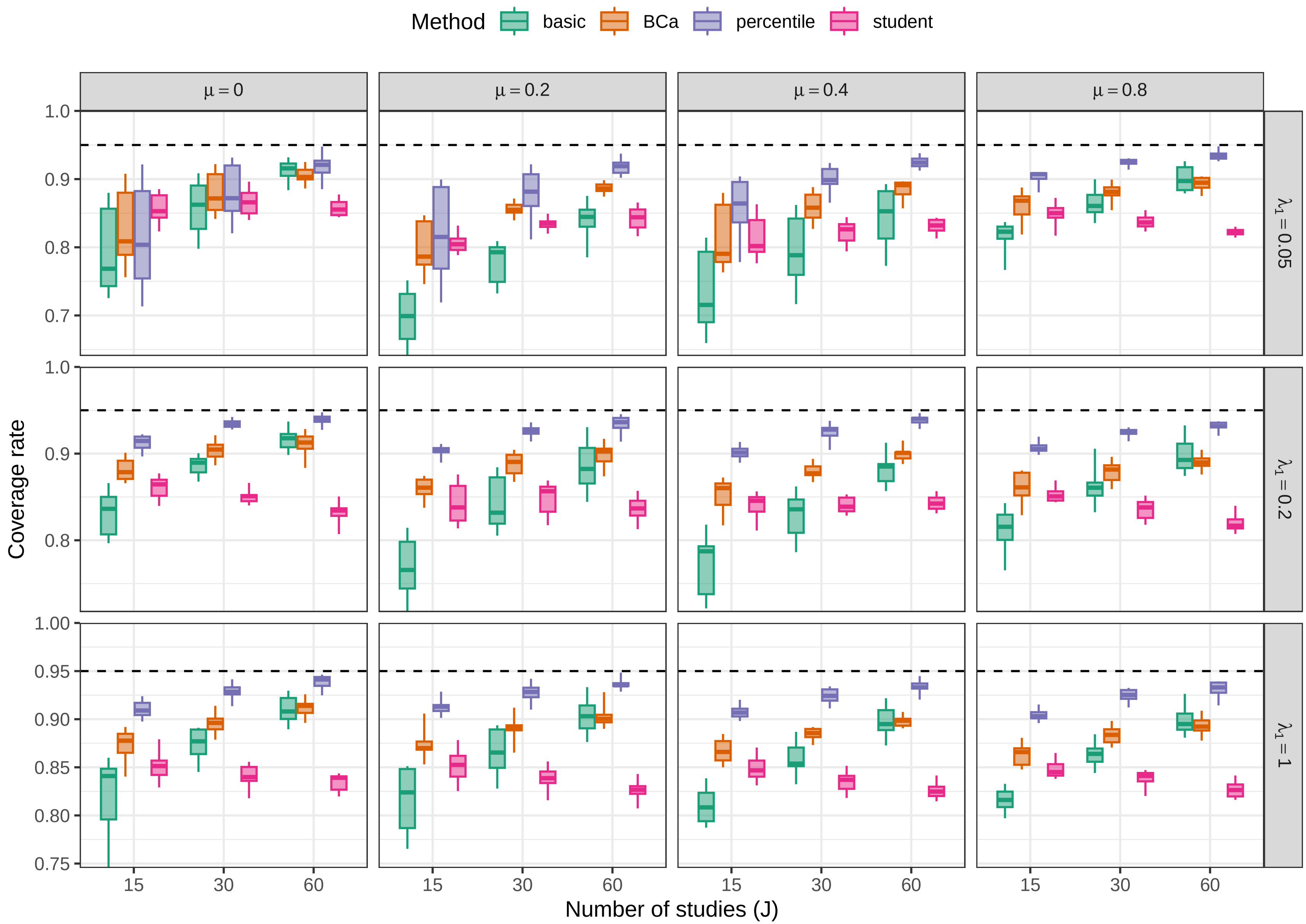

Figure D16

Coverage levels of multinomial bootstrap confidence intervals based on the PML estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a univariate selection process ($\psi = 0$). Dashed lines correspond to the nominal confidence level of 0.95.

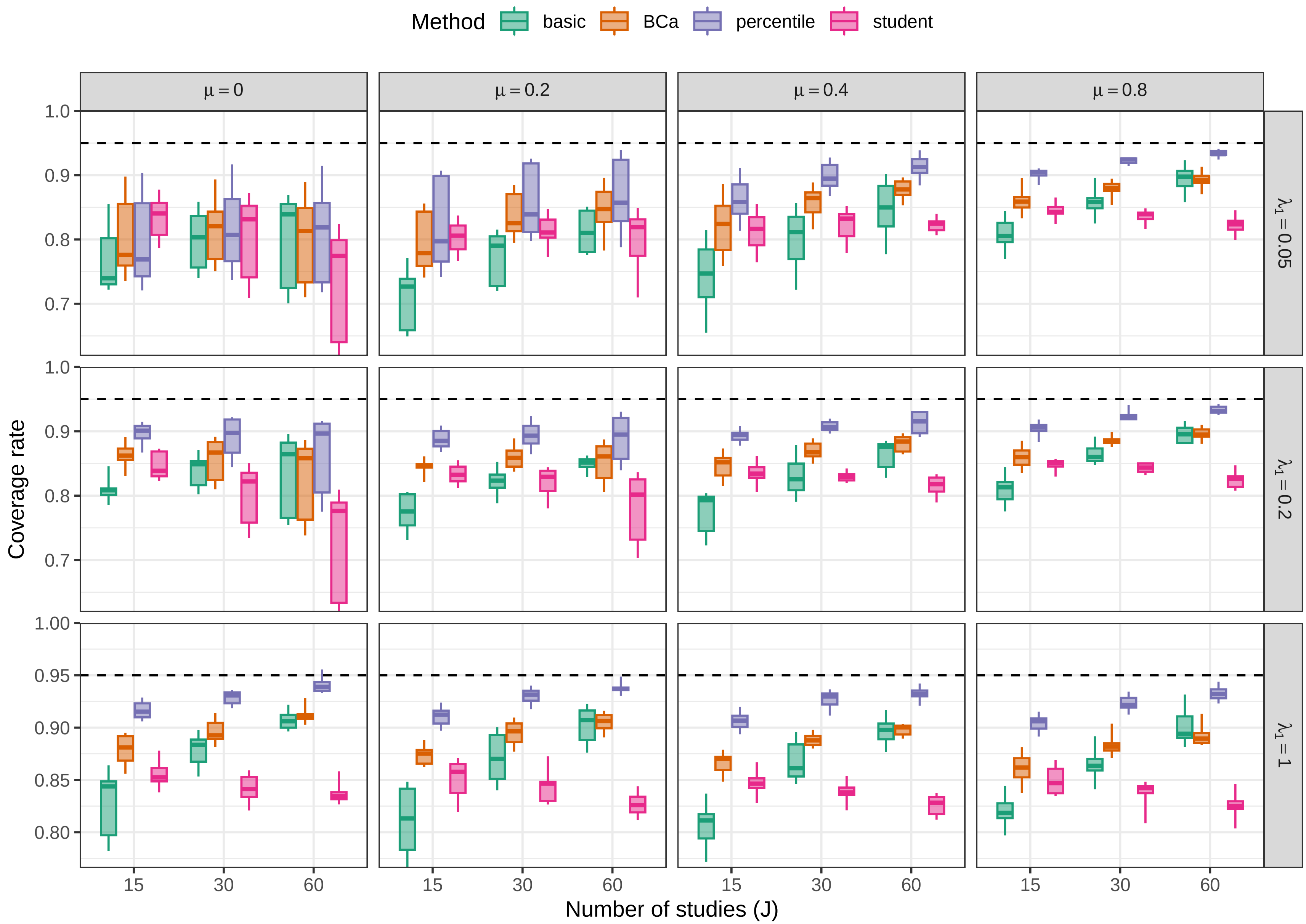

Figure D17

Coverage levels of multinomial bootstrap confidence intervals based on the PML estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a multivariate selection process ($\psi = 1$). Dashed lines correspond to the nominal confidence level of 0.95.

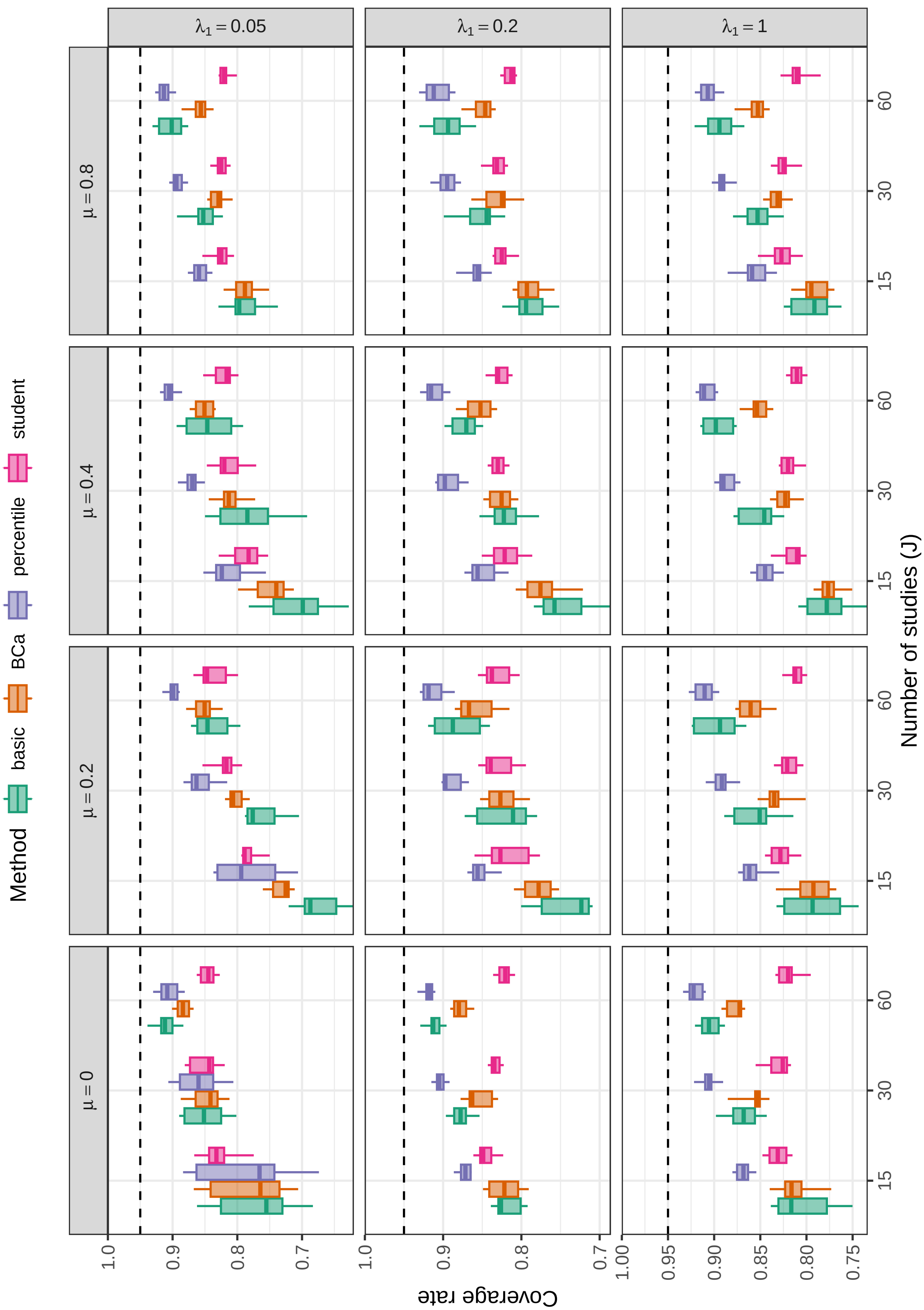

Figure D18

Coverage levels of fractional random weight bootstrap confidence intervals based on the PML estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a univariate selection process ($\psi = 0$). Dashed lines correspond to the nominal confidence level of 0.95.

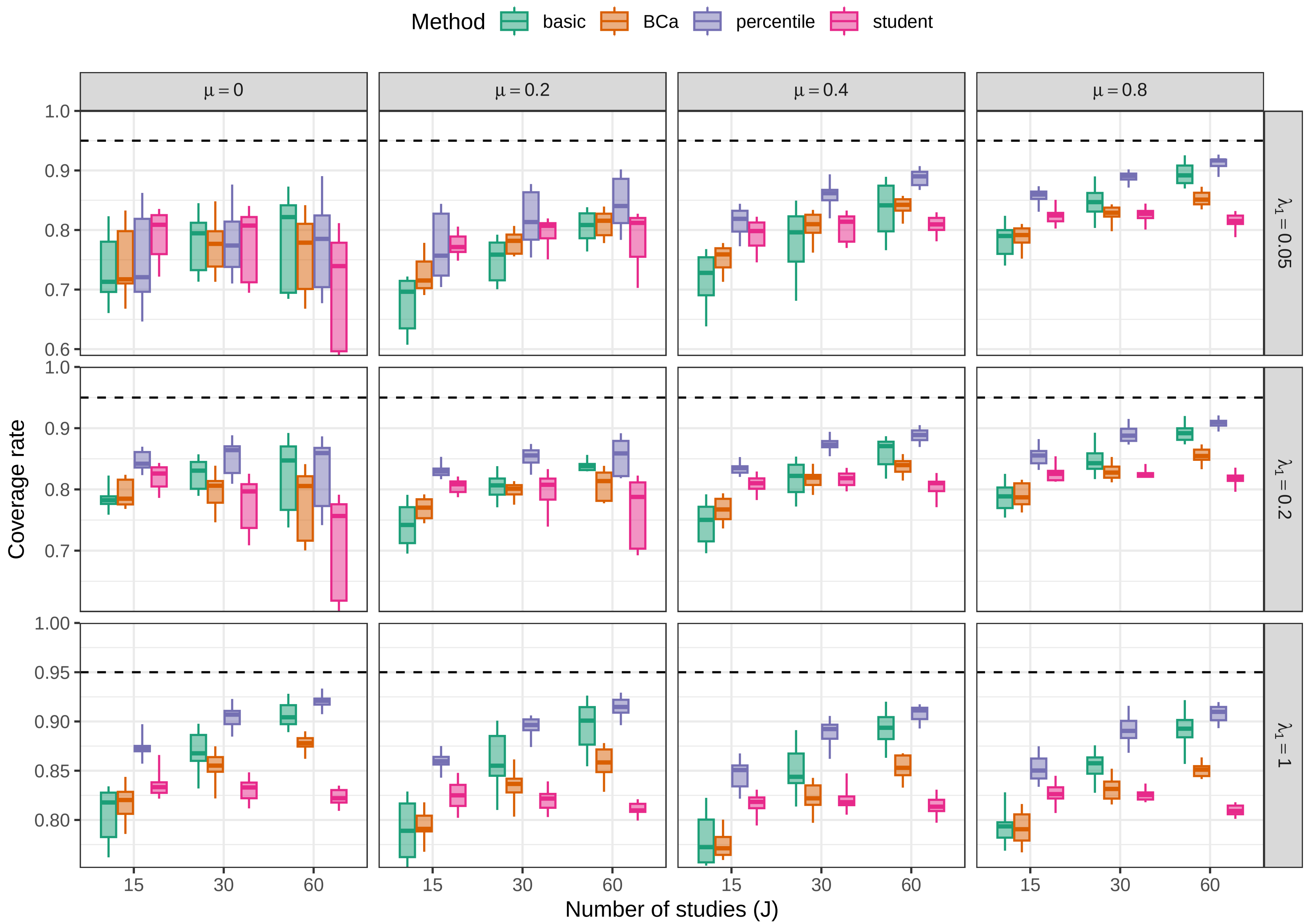

Figure D19

Coverage levels of fractional random weight bootstrap confidence intervals based on the PML estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a multivariate selection process ($\psi = 1$). Dashed lines correspond to the nominal confidence level of 0.95.

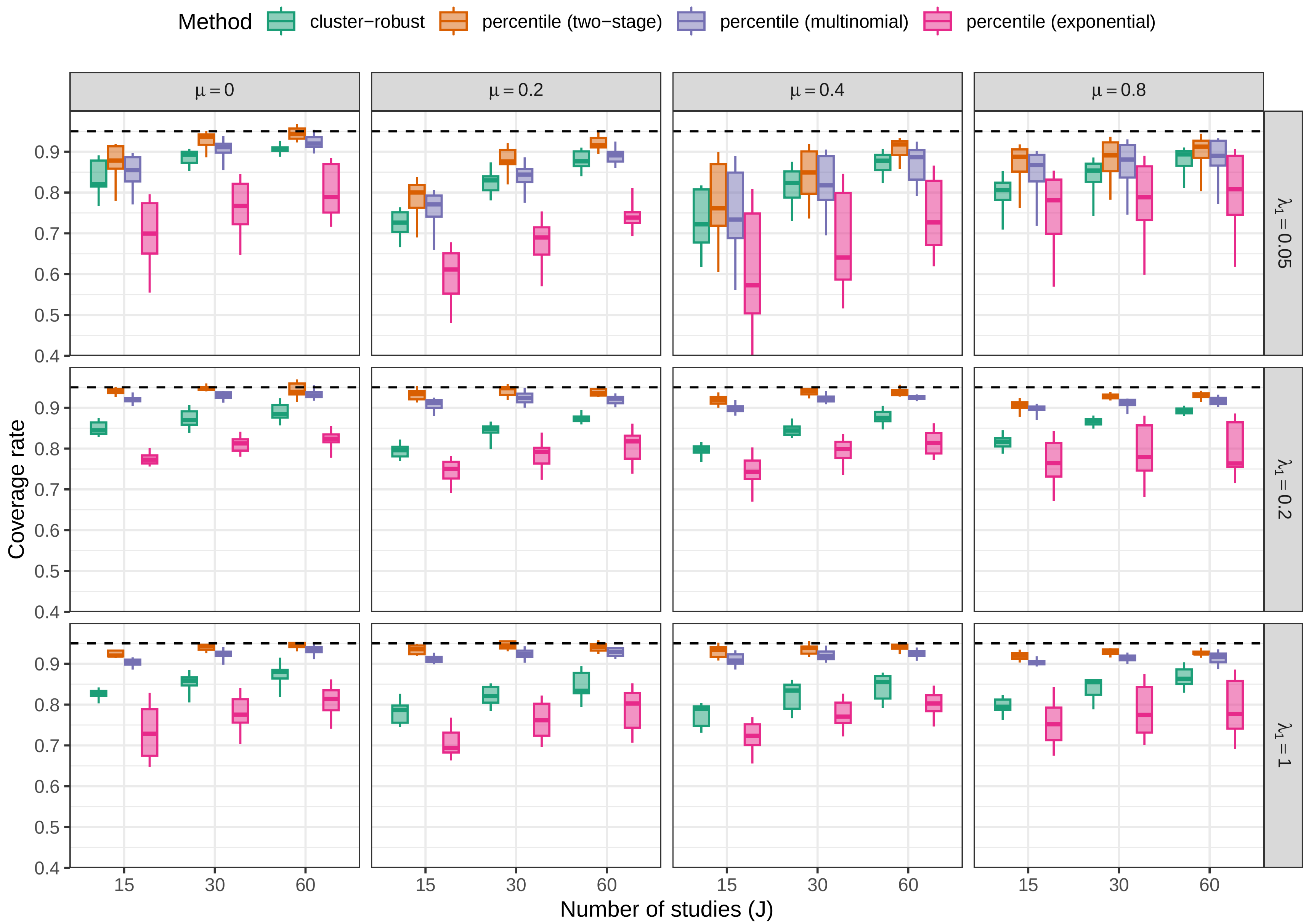

Figure D20

Coverage levels of confidence intervals based on the ARGL estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a univariate selection process ($\psi = 0$). Dashed lines correspond to the nominal confidence level of 0.95.

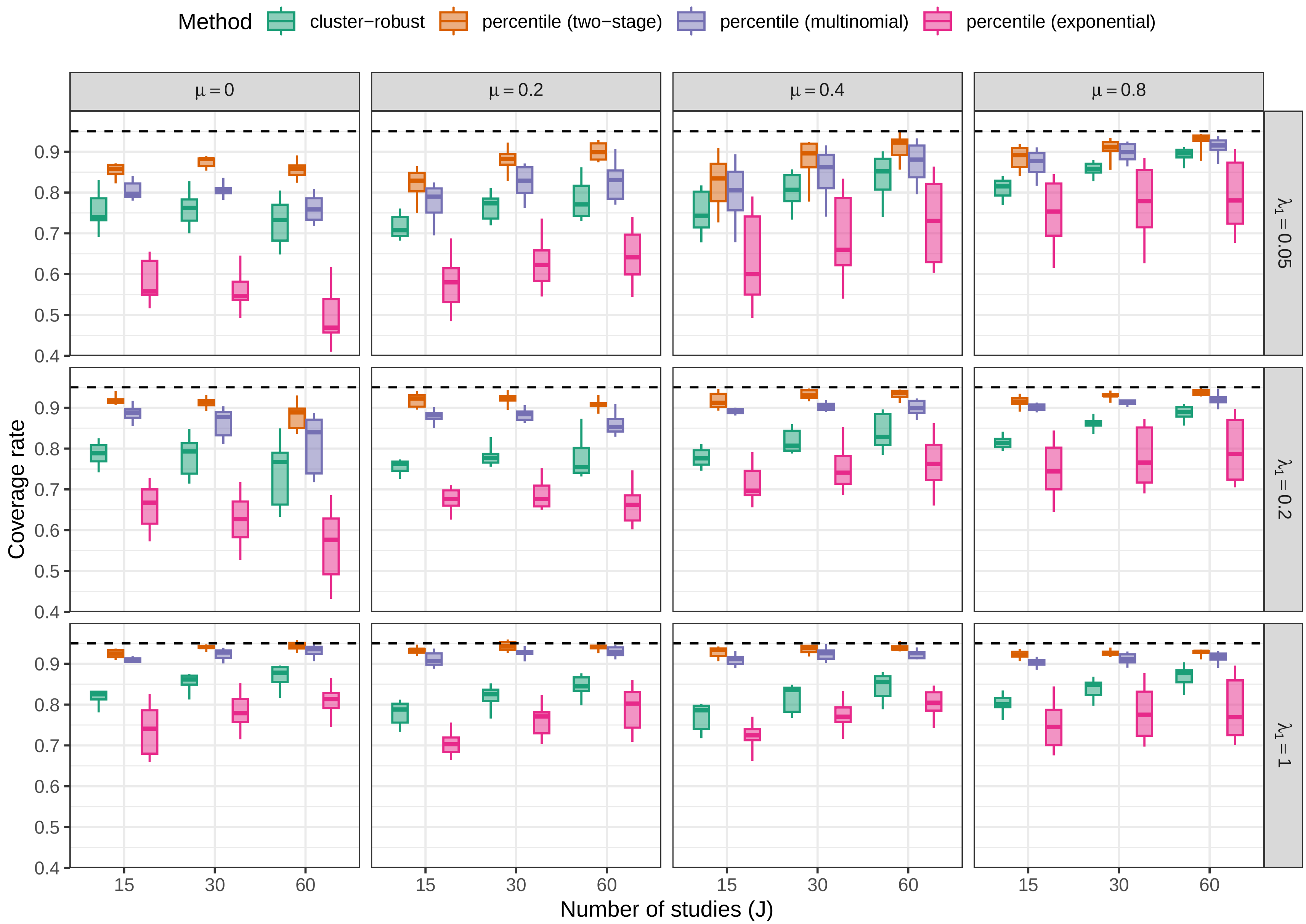

Figure D21

Coverage levels of confidence intervals based on the ARGL estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a multivariate selection process ($\psi = 1$). Dashed lines correspond to the nominal confidence level of 0.95.

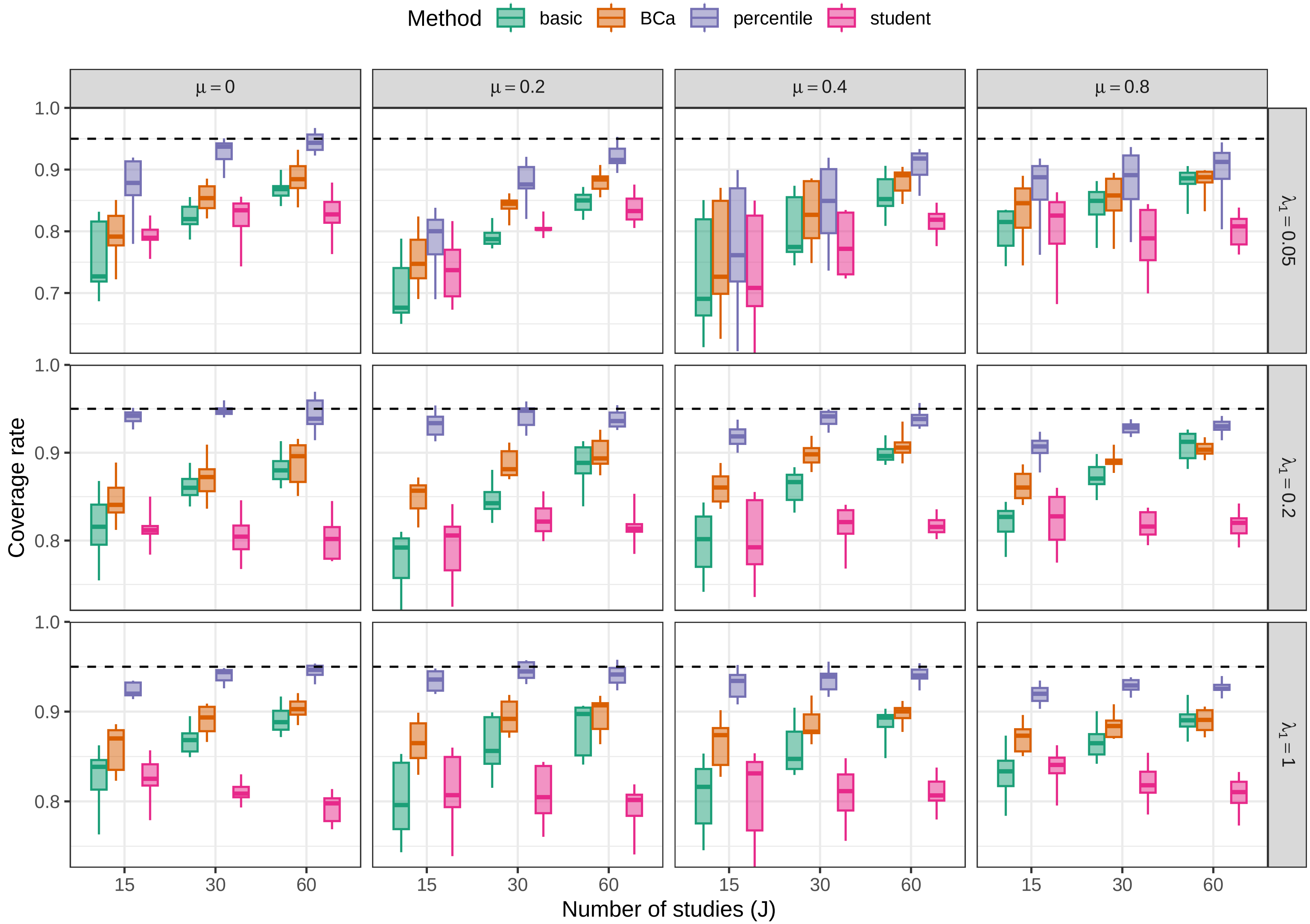

Figure D22

Coverage levels of two-stage bootstrap confidence intervals based on the ARGL estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a univariate selection process ($\psi = 0$). Dashed lines correspond to the nominal confidence level of 0.95.

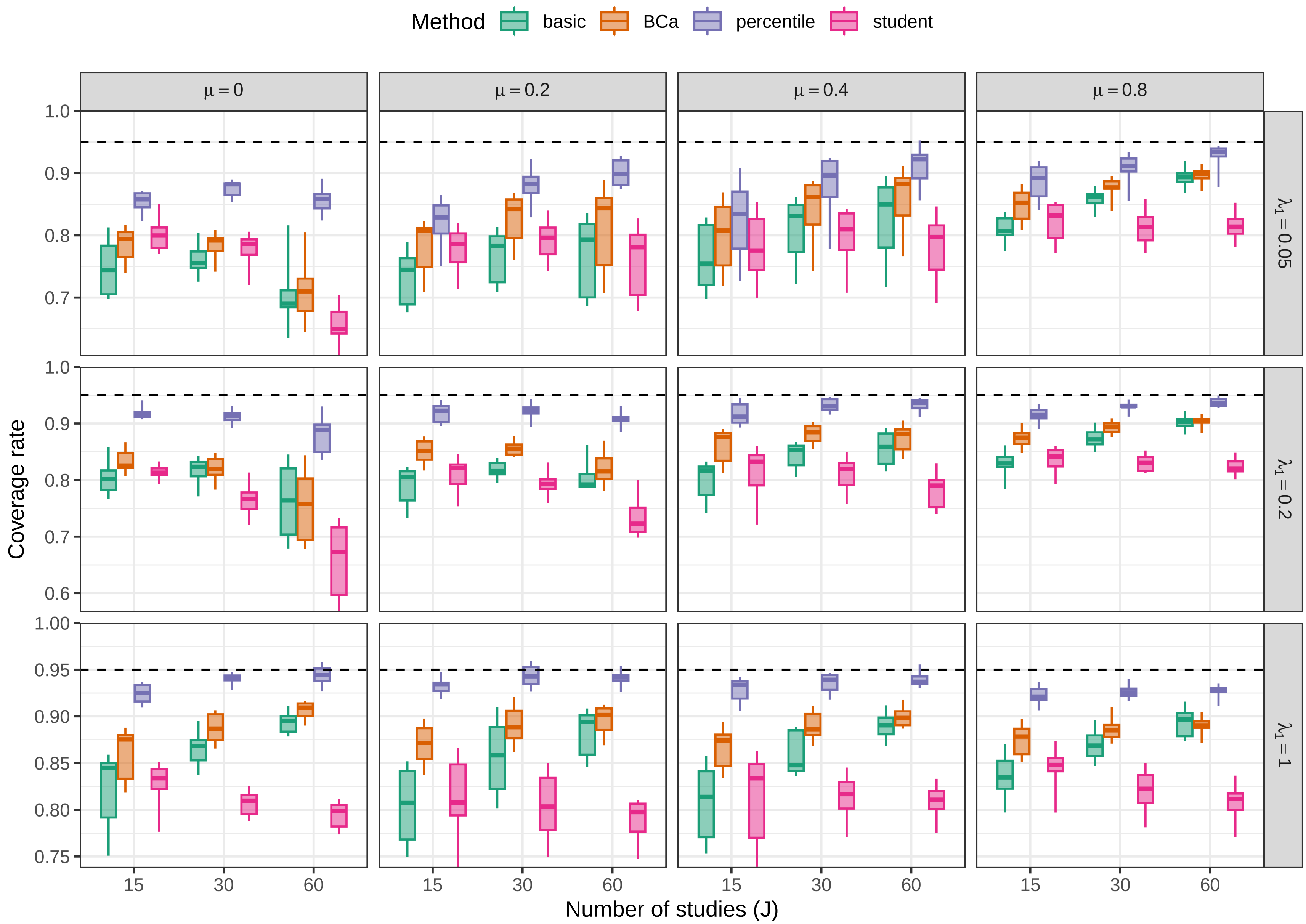

Figure D23

Coverage levels of two-stage bootstrap confidence intervals based on the ARGL estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a multivariate selection process ($\psi = 1$). Dashed lines correspond to the nominal confidence level of 0.95.

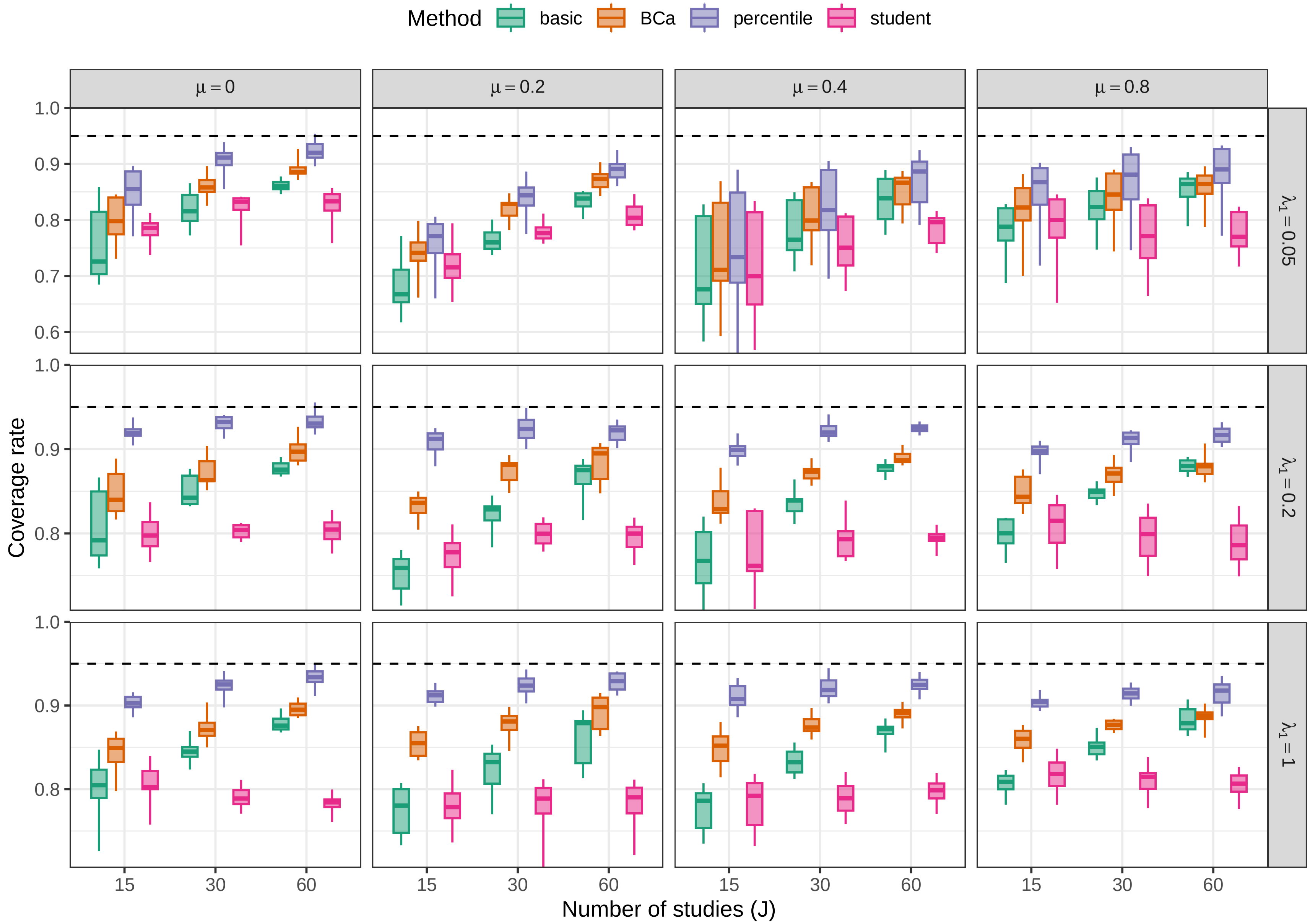

Figure D24

Coverage levels of multinomial bootstrap confidence intervals based on the ARGL estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a univariate selection process ($\psi = 0$). Dashed lines correspond to the nominal confidence level of 0.95.

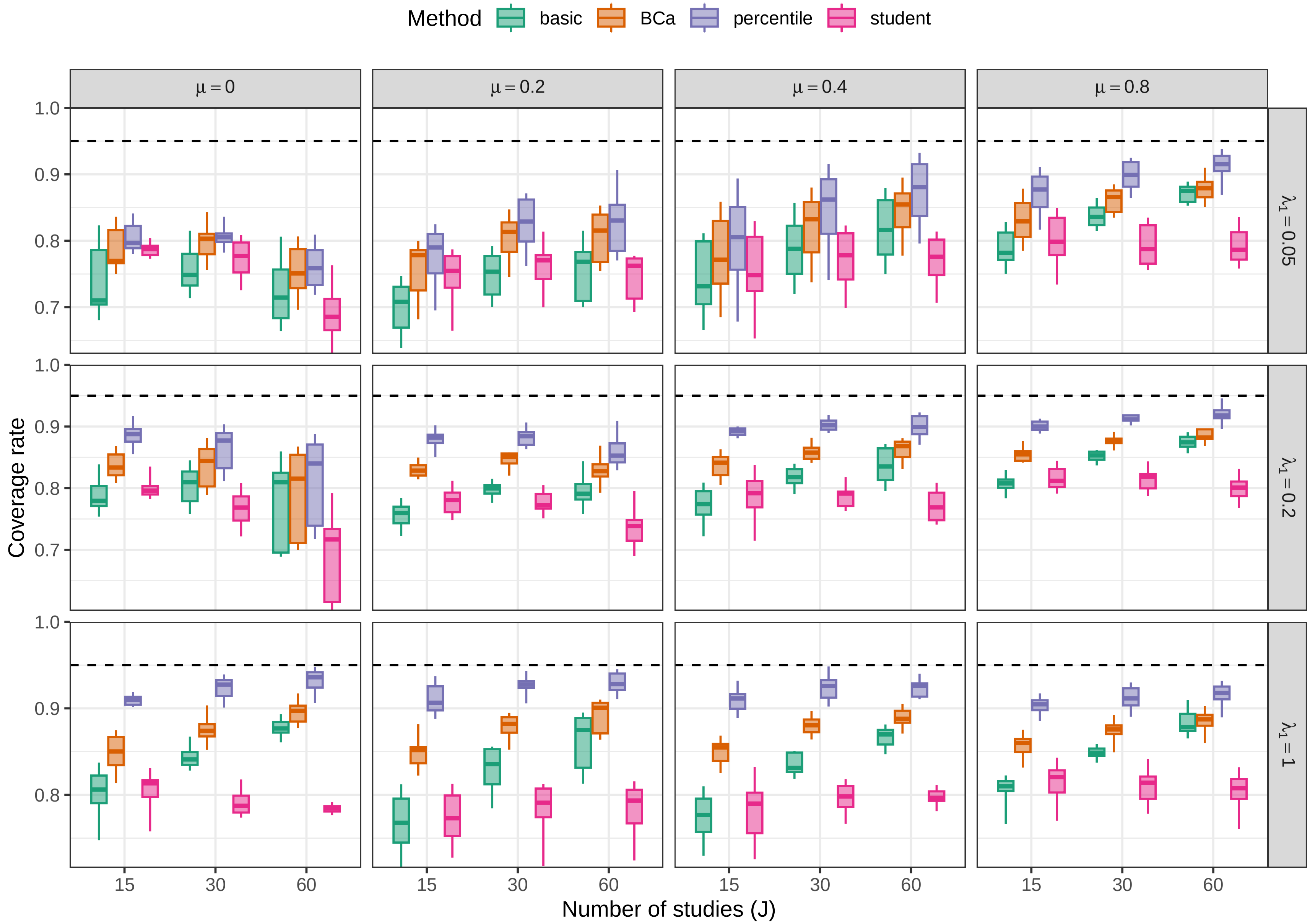

Figure D25

Coverage levels of multinomial bootstrap confidence intervals based on the ARGL estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a multivariate selection process ($\psi = 1$). Dashed lines correspond to the nominal confidence level of 0.95.

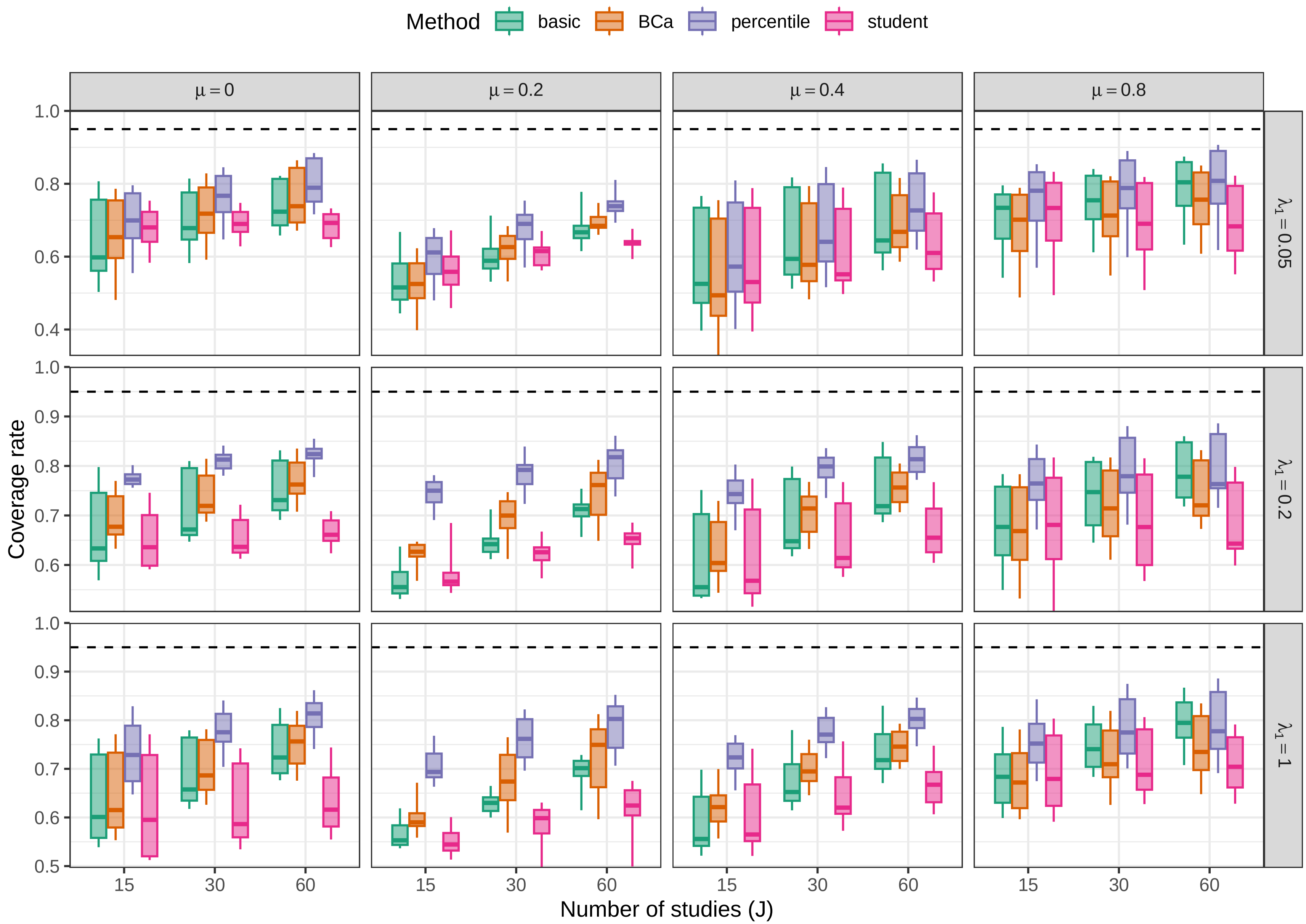

Figure D26

Coverage levels of fractional random weight bootstrap confidence intervals based on the ARGL estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a univariate selection process ($\psi = 0$). Dashed lines correspond to the nominal confidence level of 0.95.

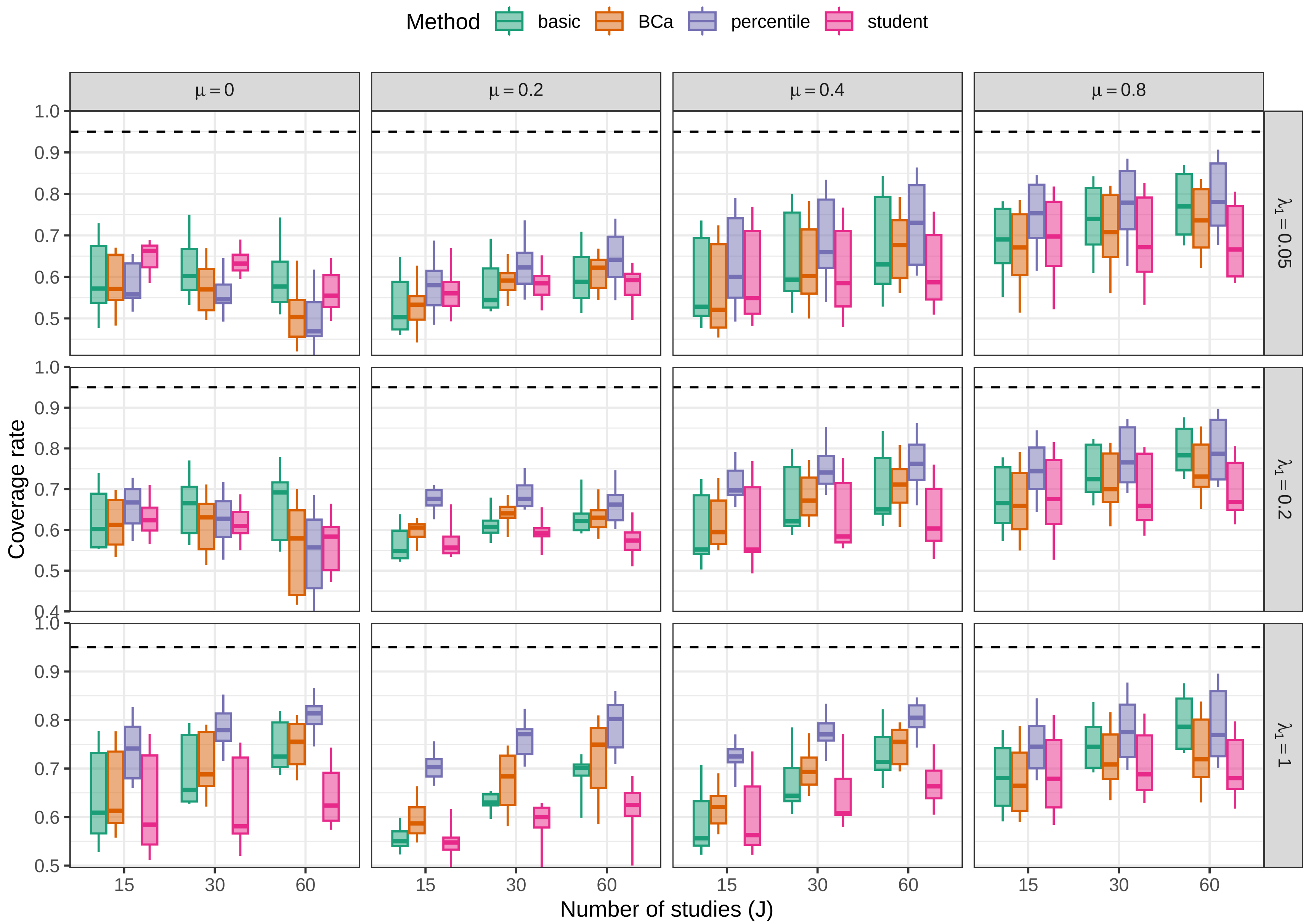

Figure D27

Coverage levels of fractional random weight bootstrap confidence intervals based on the ARGL estimator of average effect size by number of studies, average SMD, and strength of selection across scenarios with a multivariate selection process ($\psi = 1$). Dashed lines correspond to the nominal confidence level of 0.95.

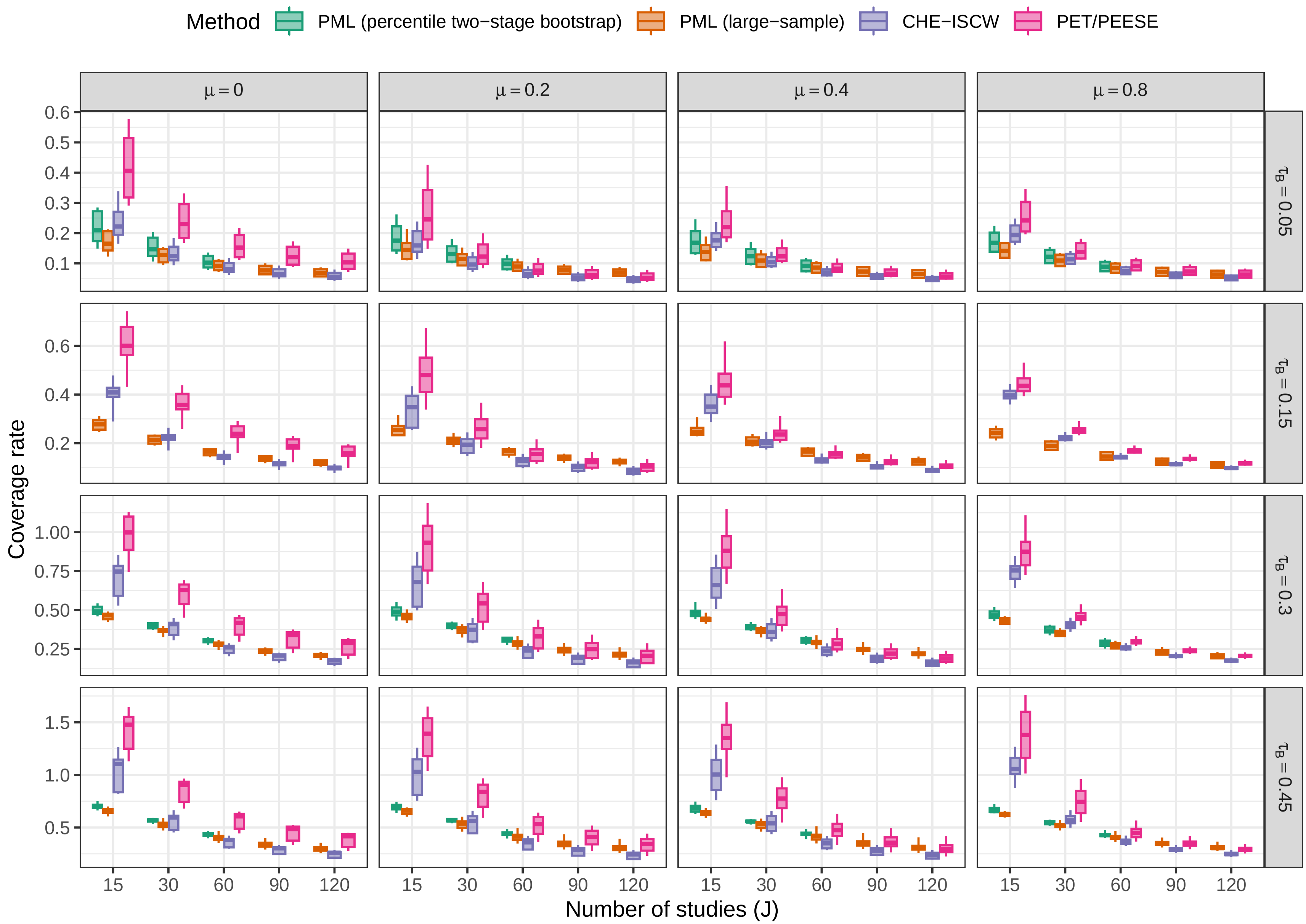

Figure D28

Average width of confidence intervals for average effect size based on selected methods, by number of studies, average SMD, and between-study heterogeneity across scenarios with a multivariate selection process ($\psi = 1$). Results are limited to scenarios with $\rho = 0.8$ and $\lambda_1 = 0.05, 0.20, 1.0$ to allow comparison between bootstrap CIs and large-sample CIs.

E Additional simulation results for estimators of heterogeneity

We briefly consider estimation of the marginal heterogeneity of the effect size distribution, for which the CHE-ISCW, PML, and ARGL methods are all relevant. Figure E1 depicts the relative bias for the CHE-ISCW, PML, and ARGL estimators of marginal variance $\tau_B^2 + \omega^2$. In most conditions, the PML and ARGL estimators are biased in the negative direction. Relative bias is high for all three estimators in conditions where the between-study heterogeneity is low ($\tau_B = 0.05$) with bias improving as between-study heterogeneity increases. The PML and ARGL estimators are less strongly biased than the CHE-ISCW estimator under conditions where there is strong selection. Neither of the selection model estimators have consistently lower bias than the other estimator.

Figure E2 depicts the relative RMSE for estimators of total heterogeneity, with Figures E3, E4, and E5 providing greater detail about the relative accuracy of the three methods. Under nearly all conditions, the PML estimator is more accurate than the ARGL estimator. The RMSE of PML is smaller than that of CHE under some but not all conditions; the former performs better under conditions where selection is strong and average SMD is low or moderate. However, just as with the estimators of average effect size, the CHE estimator is more accurate when selection is mild or absent, leading to a bias-variance trade-off.

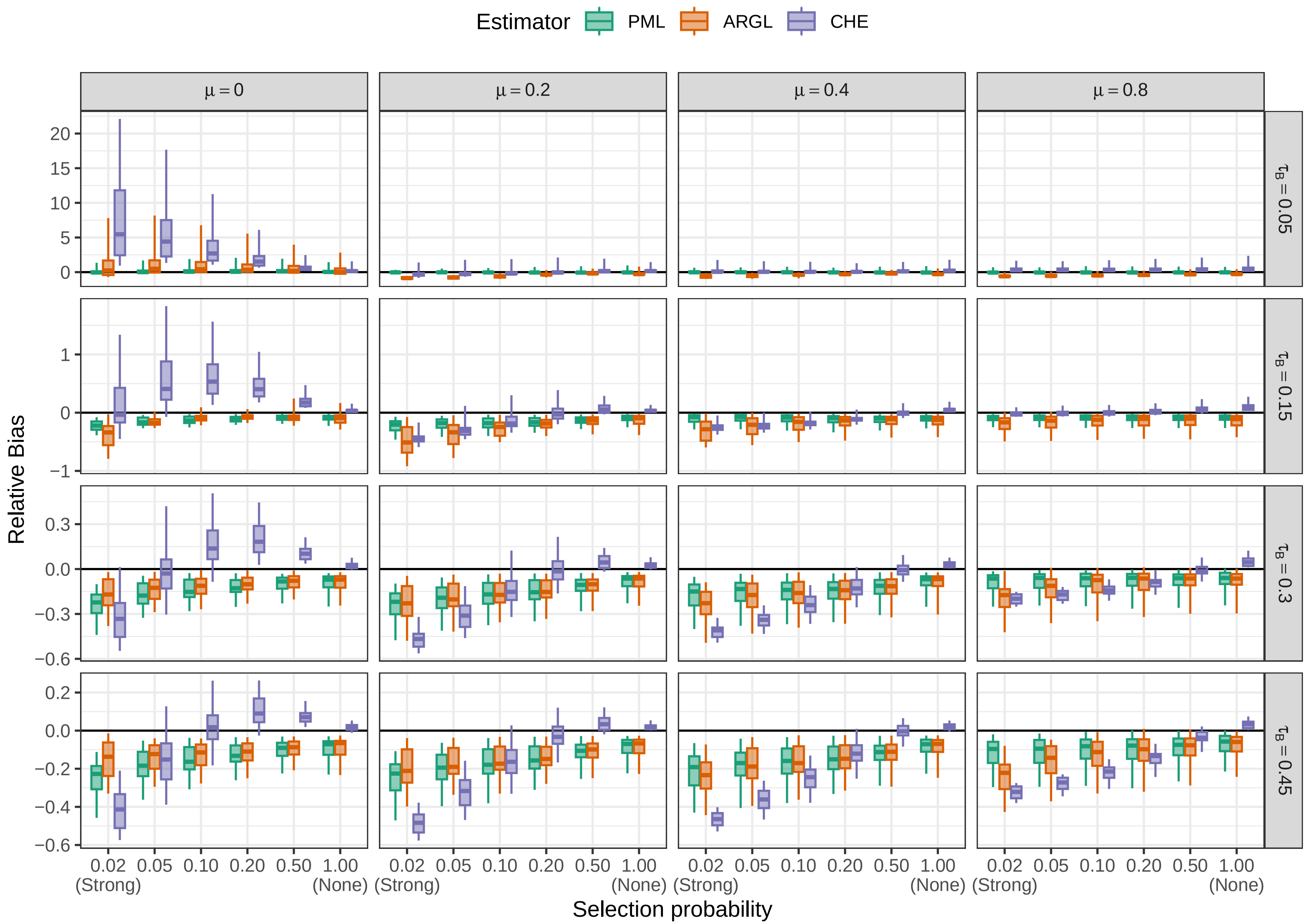

Figure E1

Relative bias of heterogeneity estimators by selection probability, average SMD, and between-study heterogeneity

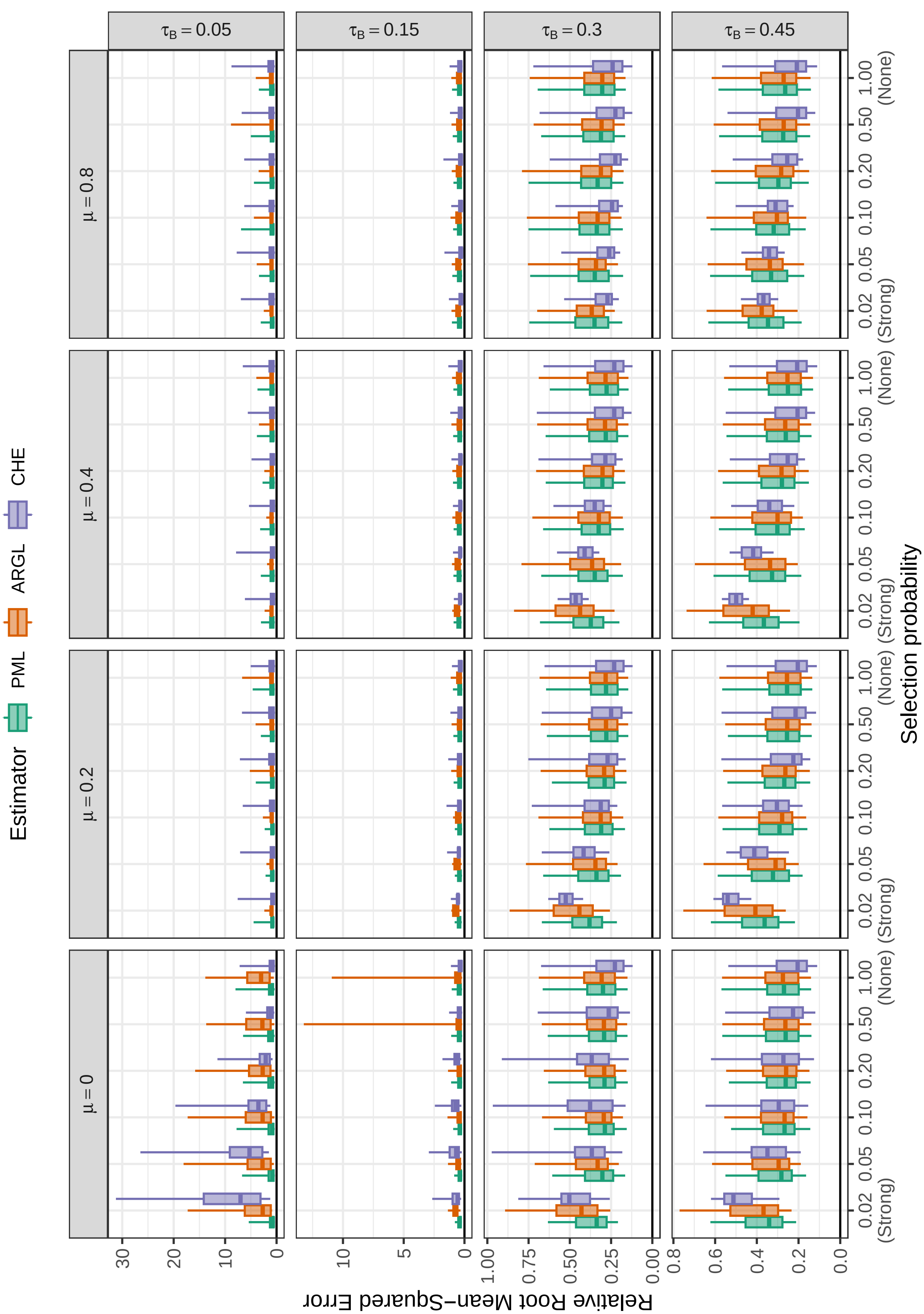

Figure E2

Relative root mean-squared error of heterogeneity estimators by selection probability, average SMD, and between-study heterogeneity

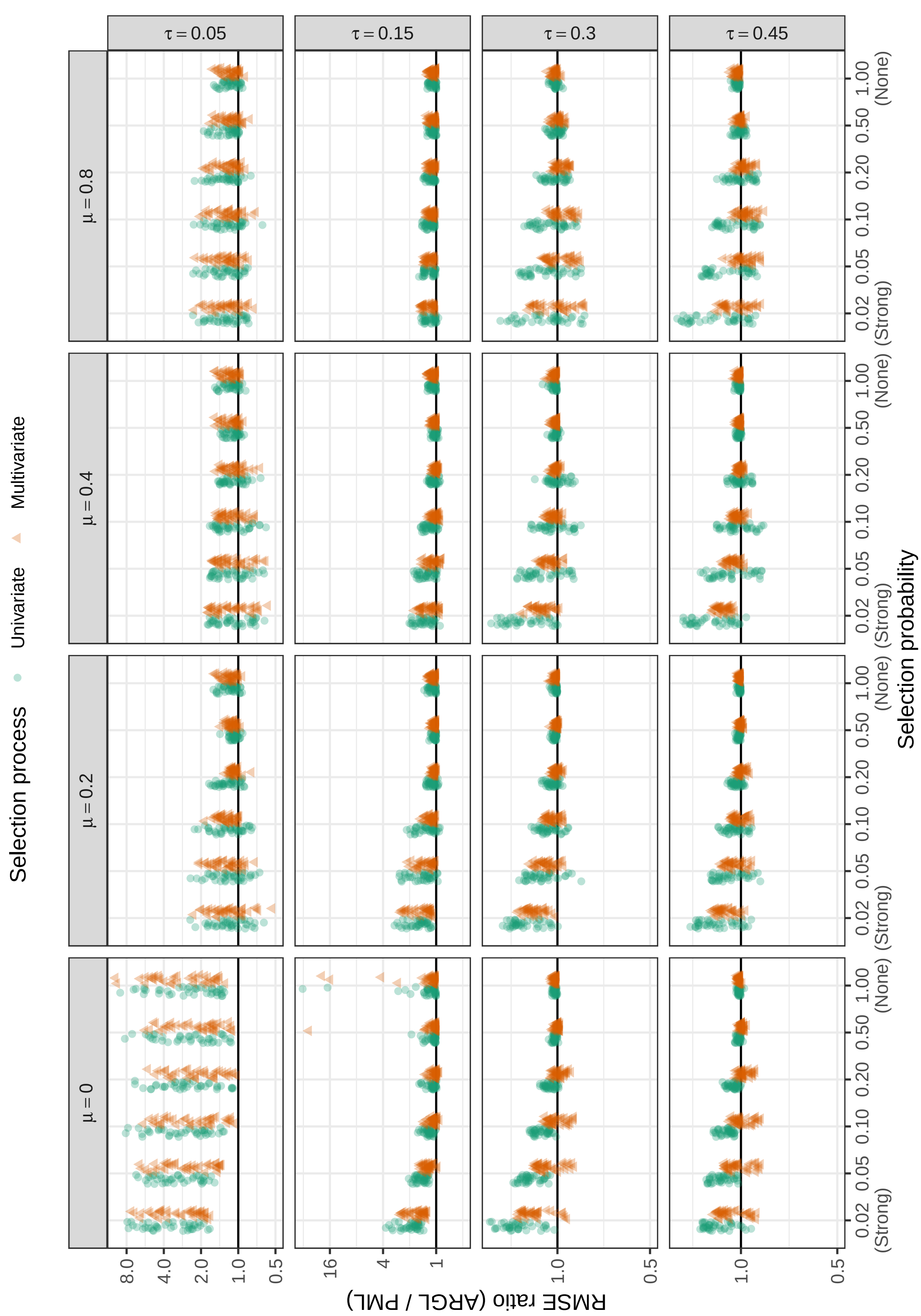

Figure E3

Ratio of root mean-squared error for ARGL heterogeneity estimator to root mean-squared error of PML heterogeneity estimator by selection process, selection probability, number of studies, average SMD, and between-study heterogeneity

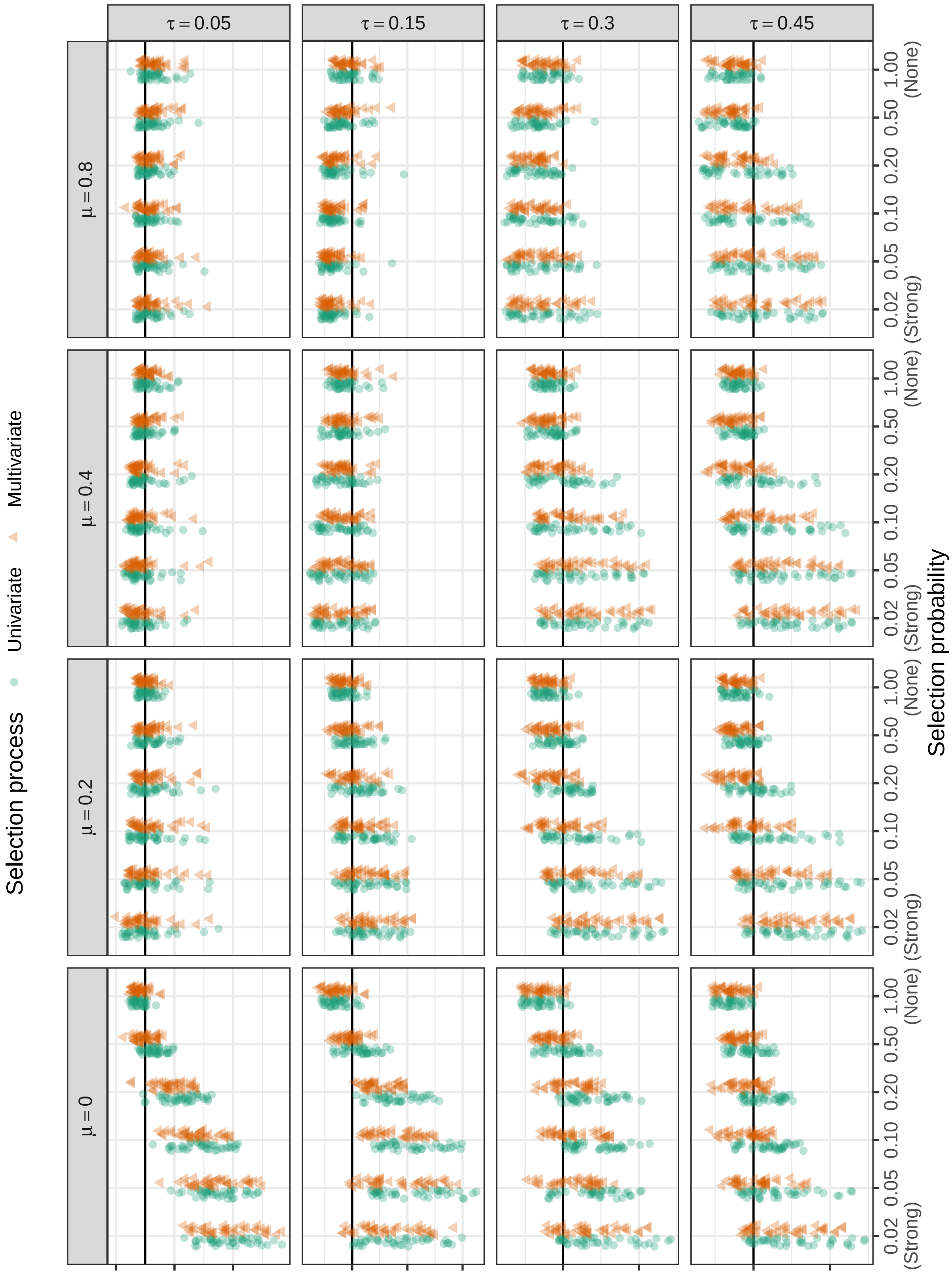

Figure E4

Ratio of root mean-squared error for PML heterogeneity estimator to root mean-squared error of CHE heterogeneity estimator by selection process, selection probability, number of studies, average SMD, and between-study heterogeneity

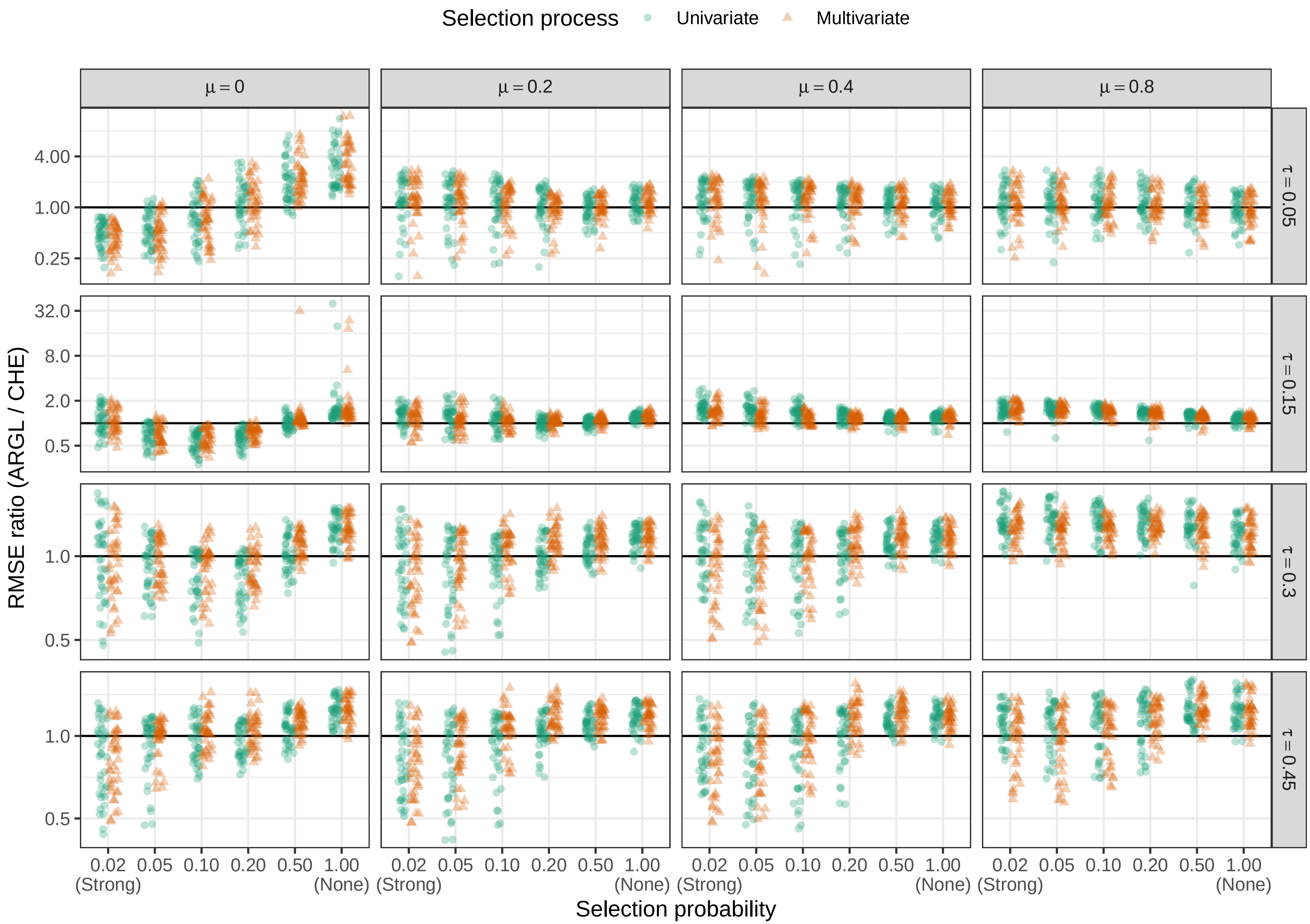

Figure E5

Ratio of root mean-squared error for ARGL heterogeneity estimator to root mean-squared error of CHE heterogeneity estimator by selection process, selection probability, number of studies, average SMD, and between-study heterogeneity

F Additional simulation results for estimators of selection parameter (λ_1)

The PML and ARGL methods both estimate the selection parameter λ_1 of the step-function model, a quantity that may be of substantive interest. Figure F1 shows that the relative bias of the two estimators is similar: Both show bias close to zero under conditions with low average SMD and high between-study heterogeneity but have positive biases under conditions with very strong selection probability, especially when average effect size is large or between-study heterogeneity is low. This pattern of bias is consistent with the specification of the prior penalties, which induce finite-sample bias when the true selection parameter is in a region with low prior probability (e.g., $\lambda_1 = 0.02$).

Figure F2 depicts the RMSE for the ARGL and PML estimators of the log-selection parameter with Figure F3 providing a more detailed view of the relative accuracy of the ARGL versus the PML estimators. The PML estimator of the selection parameter generally outperforms the ARGL estimator, especially when sample size is small.

Figures F4, F5, and F6 depict the confidence interval coverage of the two estimators with various bootstrap confidence interval estimation approaches, using two-stage clustered bootstrap resampling. Broadly, percentile intervals perform better than the other methods of constructing bootstrap confidence intervals. For the PML estimator (Figure F4), the percentile intervals have near-nominal coverage for sample sizes of $J = 30$ or larger under most conditions, except when average sample size is large. Compared to intervals based on the PML, the coverage levels of bootstrap CIs based on the ARGL estimator are generally inferior (Figure F5).

When average sample size is large, the performance of the PML percentile confidence intervals is strongly affected by the primary study sample size distribution (Figure F6). When average sample sizes are typical and $\mu = 0.8$, few non-significant effect sizes are generated, making it difficult to estimate the strength of selection. In contrast, smaller primary study sample sizes means that more are under-powered, so it is more feasible to estimate the strength of selection and construct confidence intervals with correct coverage.

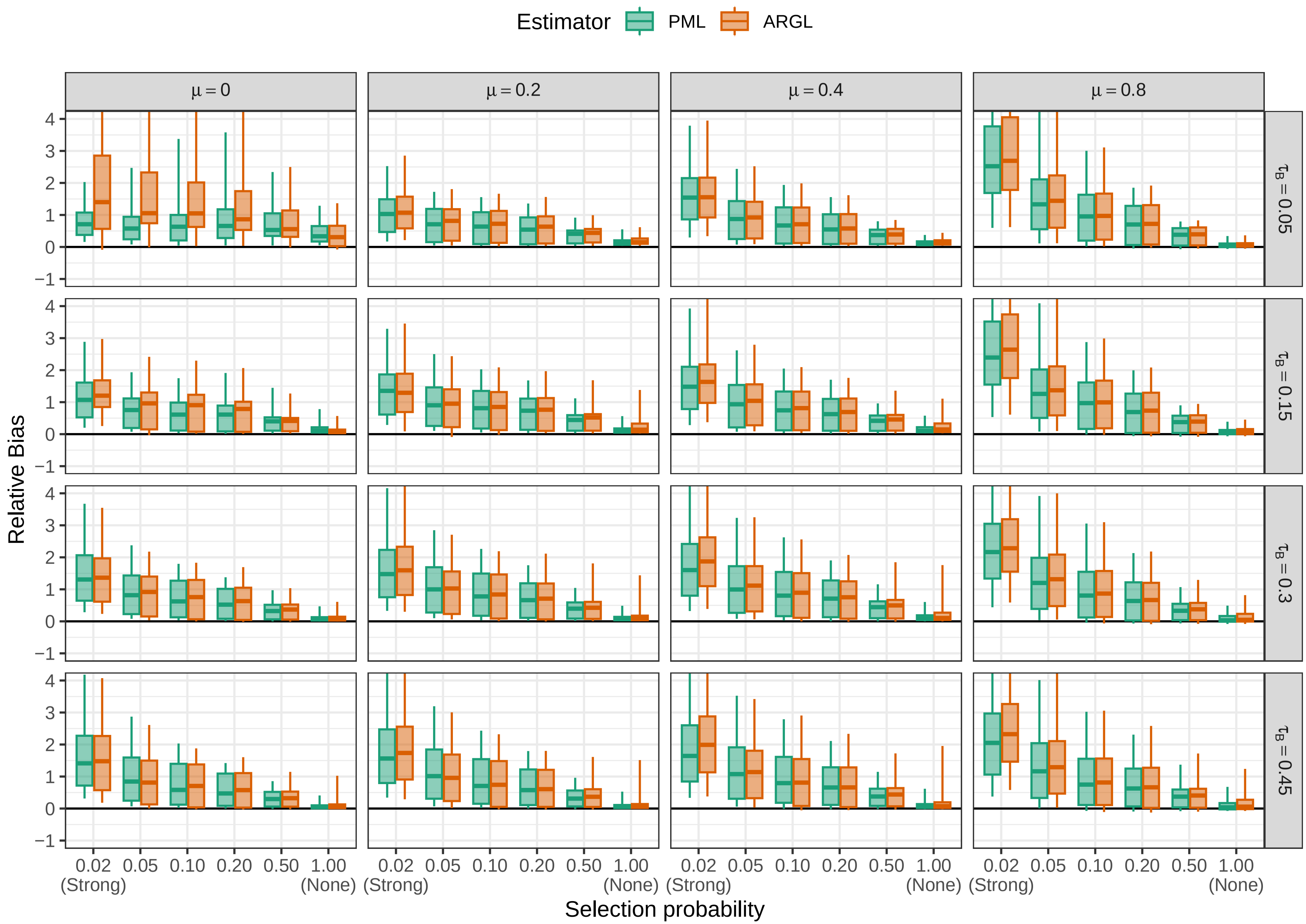

Figure F1

Relative bias for selection parameter estimators by selection probability, average SMD, and between-study heterogeneity

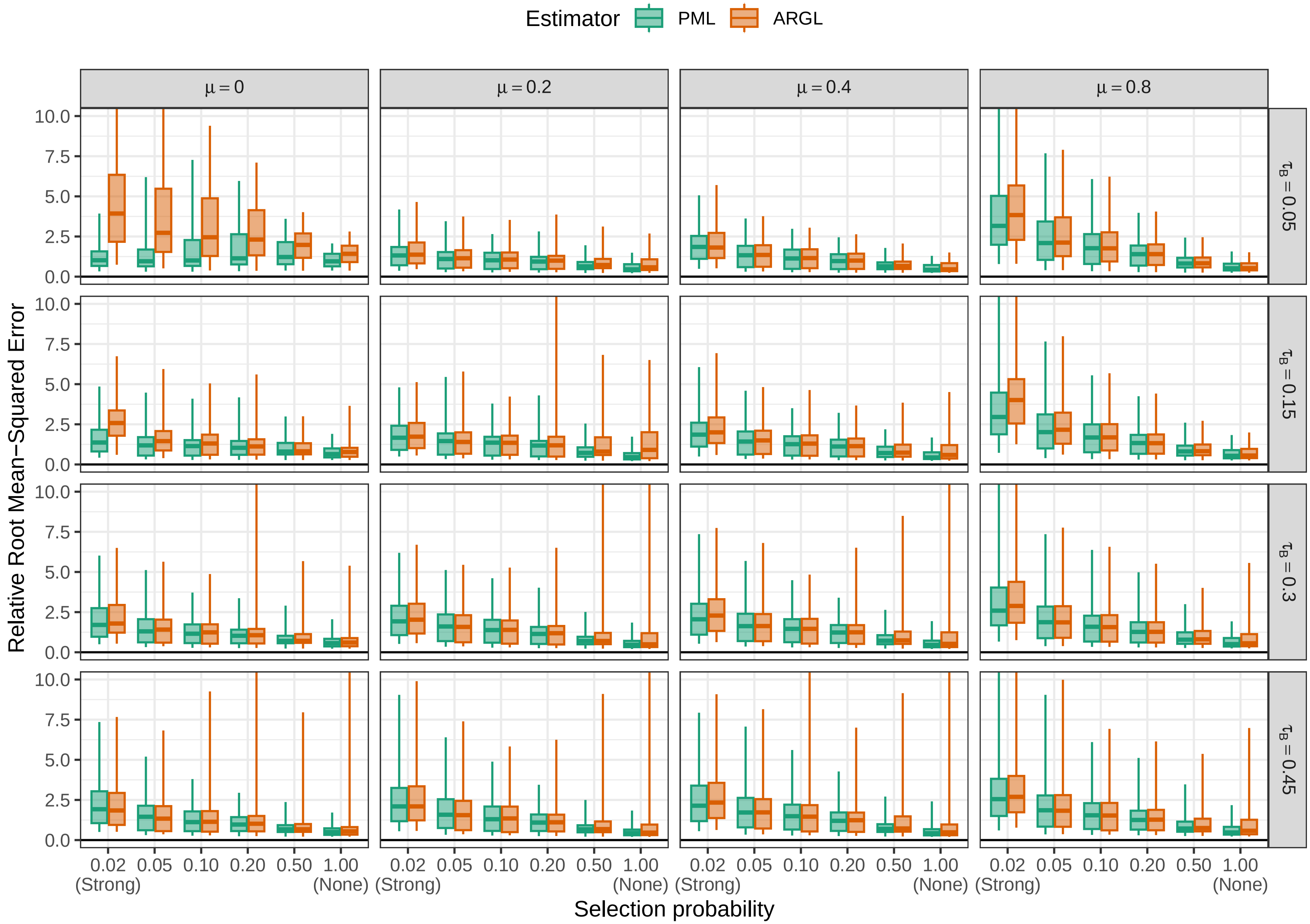

Figure F2

Relative root mean-squared error for log-selection parameter estimators by selection probability, average SMD, and between-study heterogeneity

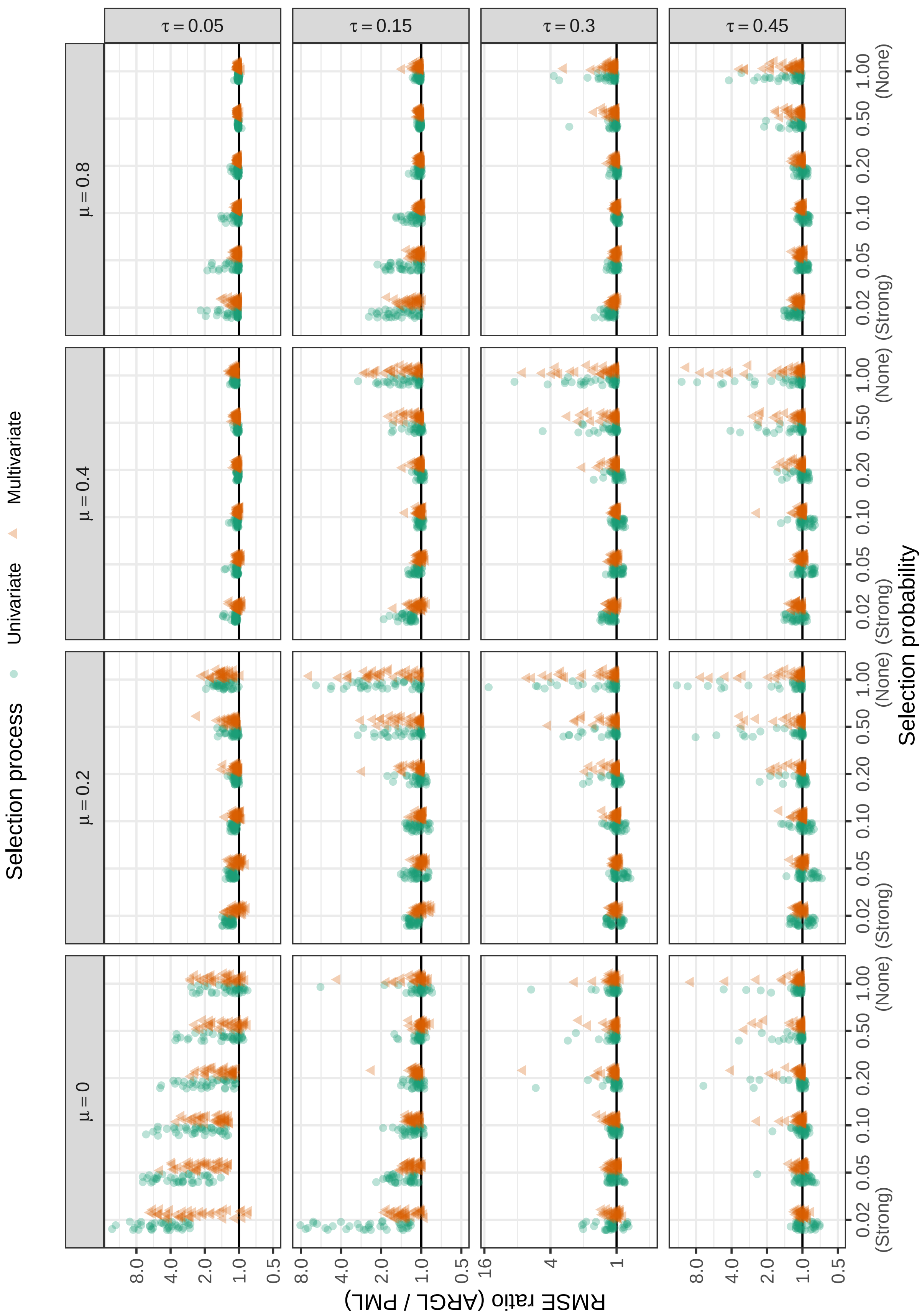

Figure F3

Ratio of root mean-squared error for ARGL selection parameter estimator to root mean-squared error of PML selection parameter estimator by selection process, selection probability, number of studies, average SMD, and between-study heterogeneity

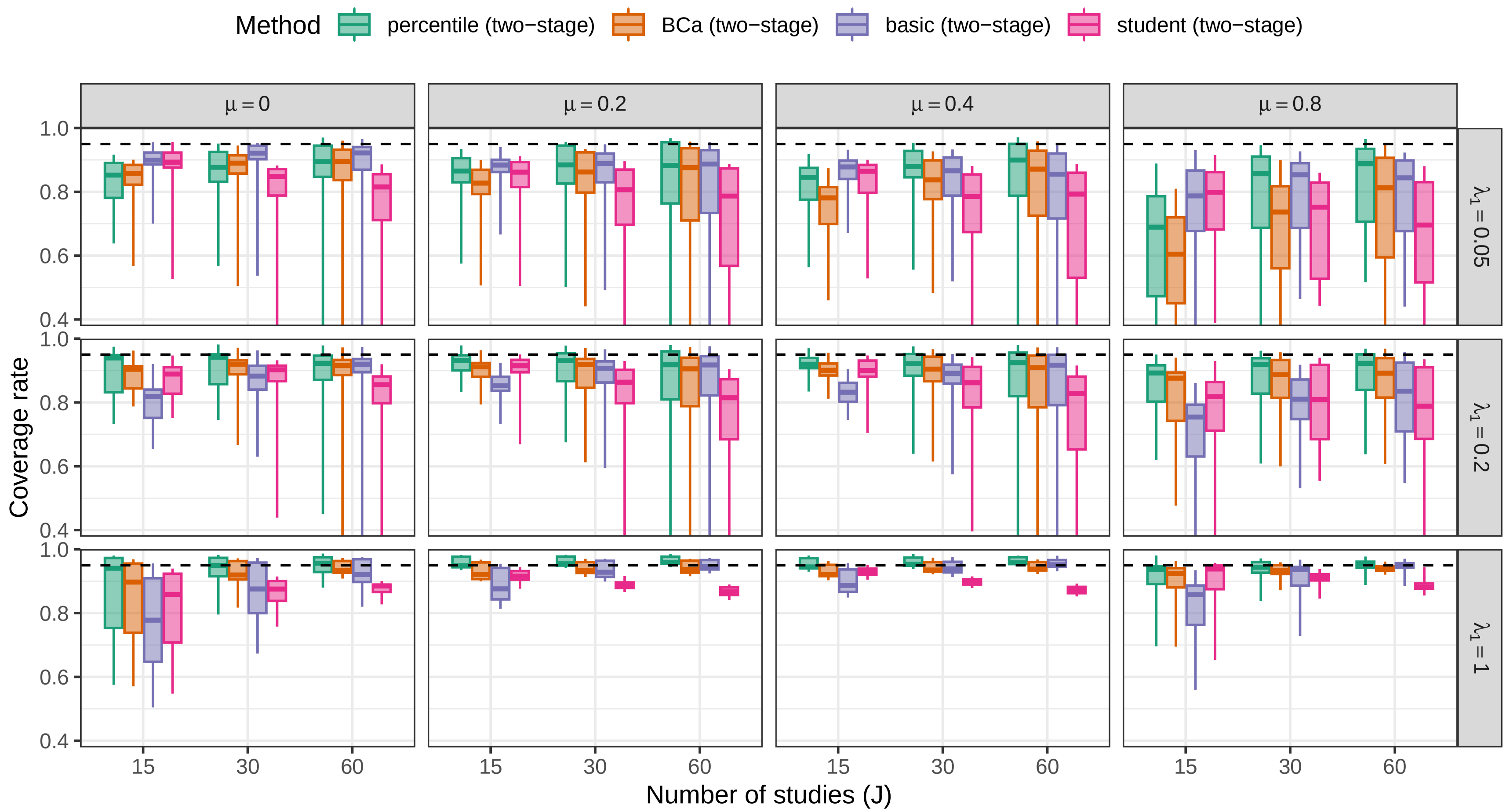

Figure F4

Coverage levels of two-stage bootstrap confidence intervals based on the PML estimator of log-selection parameter by number of studies, average SMD, and strength of selection. Dashed lines correspond to the nominal confidence level of 0.95.

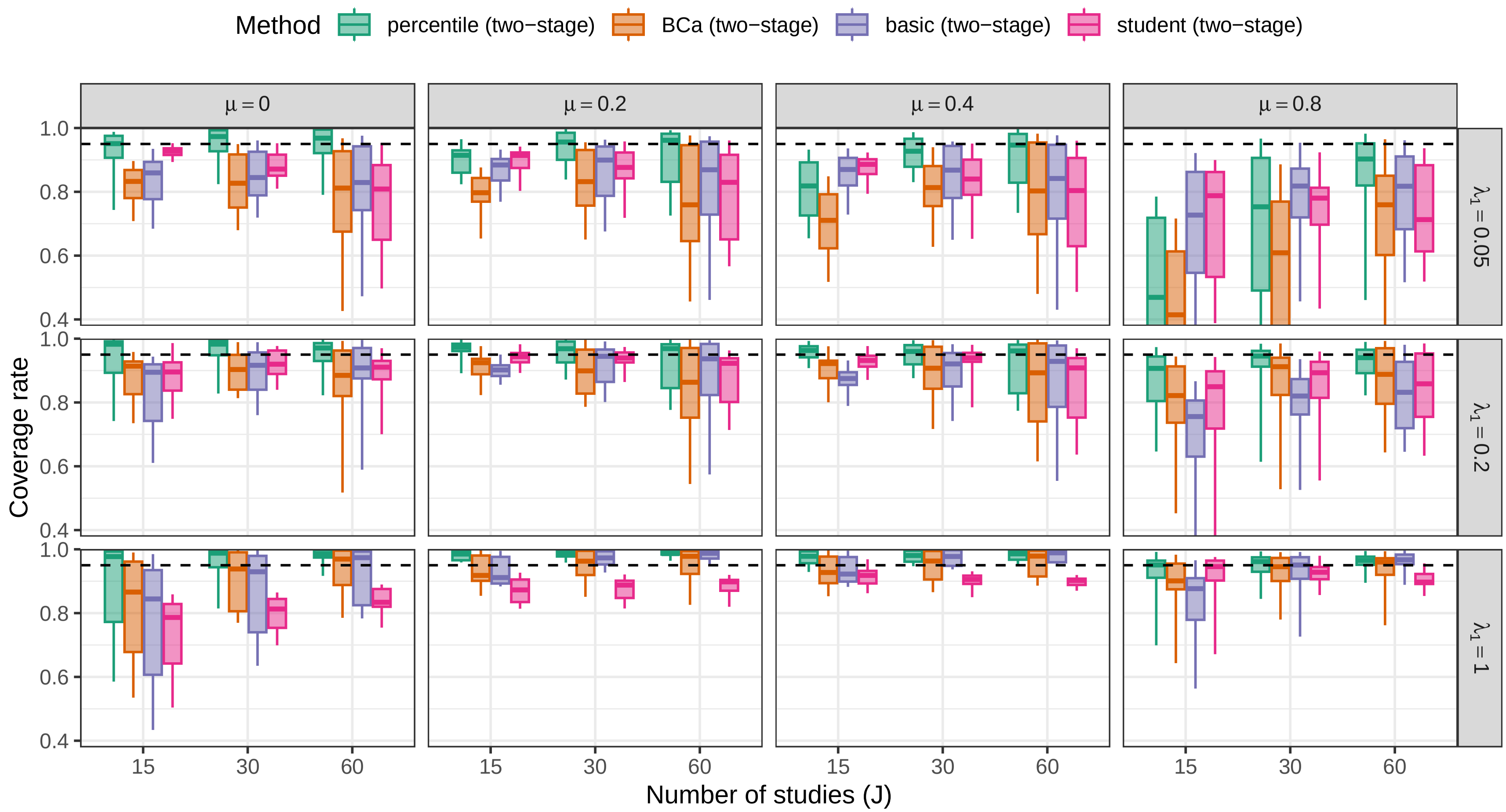

Figure F5

Coverage levels of two-stage bootstrap confidence intervals based on the ARGL estimator of log-selection parameter by number of studies, average SMD, and strength of selection. Dashed lines correspond to the nominal confidence level of 0.95.

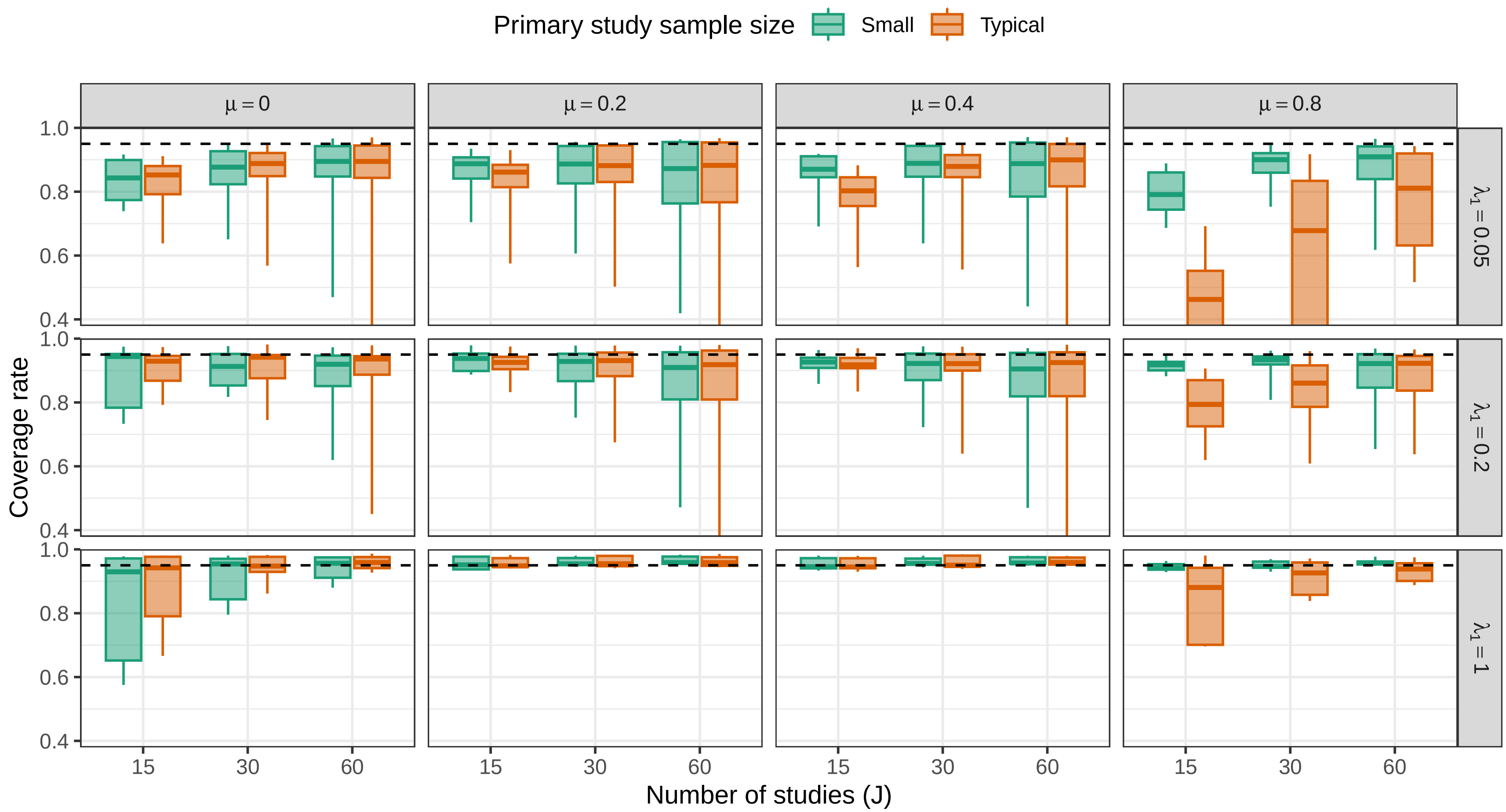

Figure F6

Coverage levels of two-stage percentile bootstrap confidence intervals based on the PML estimator of log-selection parameter by number of studies, primary study sample size, average SMD, and strength of selection. Dashed lines correspond to the nominal confidence level of 0.95.

Additional References

1. Xu L, Gotwalt C, Hong Y, King CB, Meeker WQ. Applications of the fractional-random-weight bootstrap. *The American Statistician*. 2020;74(4):345-358. doi:10.1080/00031305.2020.1731599

2. Boos DD, Zhang J. Monte carlo evaluation of resampling-based hypothesis tests. *Journal of the American Statistical Association*. 2000;95(450):486-492. doi:10.1080/01621459.2000.10474226